



Pre-trade transparency and market quality in corporate bond markets

Brian Mattmann¹

Received: 7 May 2026 / Accepted: 17 August 2026
© The Author(s) 2026

Abstract

This paper studies how pre-trade transparency is associated with market quality in corporate bond markets. We examine the Swiss corporate bond market, which features a central limit order book (CLOB) with publicly visible quotes alongside a coexisting over-the-counter (OTC) market, and compare it to the predominantly OTC-driven U.S. market. Using a comprehensive transaction-level dataset, we document that while the Swiss market is substantially smaller and exhibits lower trading activity, it displays significantly more favorable price-based liquidity measures, with lower transaction costs, price dispersion, and price impact. Importantly, these differences extend beyond CLOB-trading: even OTC transactions are executed at more competitive prices relative to the U.S. market. Moreover, OTC transaction prices closely align with publicly observable CLOB quotes: 95% occur within quoted bid-ask boundaries, most execute near the quoted midpoint. Furthermore, narrower quoted bid-ask spreads and greater quoted depth in the CLOB are associated with lower OTC transaction costs. We provide evidence consistent with the view that pre-trade transparency is associated with improved pricing by providing all market participants with a common reference point that may strengthen investors' bargaining position and encourage dealers to offer more competitive OTC prices. These findings highlight the potential role of pre-trade transparency in improving market quality in decentralized trading environments.

Keywords Corporate bonds · Pre-trade transparency · OTC · Market microstructure · Liquidity · Limit order books

JEL Classification G10 · G12 · G14 · G15 · G18 · G24

✉ Brian Mattmann
brian.mattmann@hslu.ch

¹ Lucerne University of Applied Sciences and Arts, Lucerne, Switzerland

1 Introduction

Corporate bond markets are predominantly organized as decentralized over-the-counter (OTC) markets in which investors negotiate bilaterally with dealers. Unlike equity markets, where centralized exchanges publicly display executable quotes prior to trading, most corporate bond transactions occur in opaque environments with limited pre-trade transparency. Although post-trade transparency has improved substantially following the introduction of the TRACE reporting system in the U.S., investors generally continue to lack access to publicly observable reference prices before negotiating with dealers. Corporate bond markets remain characterized by search frictions, dealer intermediation rents, and relatively high transaction costs.

This paper studies whether pre-trade transparency is associated with improved market quality in corporate bond markets. We examine the Swiss corporate bond market, where publicly observable executable quotes are available through a central limit order book (CLOB) while an OTC market continues to coexist alongside the exchange, particularly for larger transactions. This institutional setting allows us to study not only the role of transparency in centralized trading, but also whether publicly observable quotes influence pricing outcomes in decentralized OTC negotiations. Using a comprehensive transaction-level dataset covering all Swiss franc-denominated corporate bond transactions in 2018, combined with transaction-level data from TRACE, we document several striking differences between the Swiss and U.S. corporate bond markets. Although the Swiss market is substantially smaller and exhibits lower trading activity, it displays significantly more favorable price-based liquidity characteristics. Transaction costs, price dispersion, and price impact are all substantially lower in Switzerland than in the U.S. market, even after controlling for bond characteristics such as issue size, maturity, age, and credit quality.

Importantly, these differences are not limited to exchange-based trading. OTC transactions in Switzerland are also executed at substantially more competitive prices than comparable OTC transactions in the U.S. Moreover, OTC transaction prices closely align with publicly observable quotes displayed in the CLOB: 95% of OTC transactions occur within the quoted bid-ask range observed in the CLOB, while most executions take place close to quoted midpoint prices. These findings suggest that publicly observable pre-trade prices may serve as economically meaningful reference points in bilateral dealer negotiations. We interpret the evidence as being consistent with the view that pre-trade transparency improves investors' bargaining positions in decentralized markets. When investors can observe executable quotes prior to negotiation, dealers may face stronger competitive pressure to offer favorable pricing, reducing their ability to extract informational rents from less informed clients. In this sense, the benefits of transparency may extend beyond centralized trading venues and influence pricing outcomes even in dealer-intermediated OTC markets. By contrast, transaction costs for electronically executed trades through CLOB-type trading protocols are broadly comparable across the two markets. This finding is notable because it suggests that pricing outcomes converge when similar electronic trading protocols are used.

A large literature studies the effects of transparency in corporate bond markets and documents that the dissemination of transaction information post-trading improves

market quality and reduces trading costs (e.g., Bessembinder et al. 2006; Edwards et al. 2007; Goldstein et al. 2007). At the same time, important frictions remain: Transaction costs continue to vary substantially across investors and trade sizes, with smaller and less active investors often receiving systematically worse prices (Schultz 2001; O'Hara et al. 2018; Hendershott et al. 2020). Recent evidence further suggests that dealer intermediation increasingly relies on matching customer order flow rather than committing balance sheet capital (Bessembinder et al. 2018; Schultz 2017; Choi et al. 2024; Craig et al. 2025). These findings raise the question of whether broader access to pre-trade price information could improve competition and market quality in decentralized corporate bond markets. Despite its importance for market design and regulation, empirical evidence on the role of pre-trade transparency in corporate bond markets remains limited. The main challenge is that most corporate bond markets operate without publicly observable executable quotes prior to trading. Existing studies therefore typically rely on subsets of exchange-listed bonds or specific electronic trading platforms. For example, Chen and Zhong (2017) show that New York Stock Exchange (NYSE)-listed corporate bonds exhibit lower bid-ask spread dispersion and lower transaction costs, while Holden et al. (2018) document that exchange-listed bonds incorporate information more efficiently. Abudy and Wohl (2018) compare the centralized Israeli bond market to the U.S. market and argue that centralized order book trading is associated with lower transaction costs and greater price competitiveness. Our paper contributes to the literature in several ways: First, we provide direct empirical evidence on the relationship between pre-trade transparency and market quality using a real-world market structure in which a transparent exchange coexists with an OTC market. Second, we show that the potential benefits of transparency are not limited to exchange executions, but may also spill over into decentralized OTC trading. Third, by comparing markets with substantially different transparency regimes, we complement existing studies focused on individual platforms or subsets of exchange-listed bonds. More broadly, our findings contribute to ongoing discussions regarding market design and transparency in corporate bond markets.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature on corporate bond trading, discusses the underlying economic mechanisms, and develops the hypotheses. Section 3 provides a market description of the U.S. and Swiss corporate bond markets and introduces the data sets. Section 4 studies differences in quantity- and price-based liquidity measures across the two markets. Section 5 demonstrates the differences in market designs before examining the benefits of a public CLOB for market participants and the quality of the market. Section 6 concludes the paper by discussing the importance of the results for future market structure reforms in corporate bond trading.

2 Related literature and hypotheses development

2.1 Related literature

A number of papers have studied liquidity and transaction costs in U.S. corporate bond markets, where dealers provide liquidity to their customers on an RFQ basis. The most salient results in the literature are that liquidity is scarce and transaction costs are high, especially for smaller orders traded by retail investors; for example, see Schultz (2001) and Edwards et al. (2007). Moreover, Green et al. (2007) find that dealers are able to exercise substantial market power over investors and that this power decreases with trade size. The model of Lee and Wang (2025) implies that the dealers price discriminate by favoring likely-uninformed traders in OTC trading. O'Hara et al. (2018) as well as Hendershott et al. (2020) have demonstrated that even less active institutional investors receive relatively worse prices. This suggests that dealers are able to exploit their bargaining power and collect rents from less informed investors in an opaque OTC trading environment.

An additional driver that affects liquidity and transaction costs in corporate bond trading is the change in dealers' risk capacity since the financial crisis, which has prompted dealers to adjust their trading strategy. Today, liquidity providers are more reluctant to commit capital to customers and instead more often broker trades directly between investors on riskless terms, as documented by Harris (2015); Goldstein and Hotchkiss (2020); Bessembinder et al. (2018); Schultz (2017); and Choi et al. (2024). Dealers are able to generate risk-free profits simply by matching orders across their customer networks. Craig et al. (2025) find that in 2023 Q4, 42% of all customer trades appear to be pairing trades, with 80% of a pair trade's legs being executed in less than one second. This represents a substantial increase from 16% in 2005 Q1. Particularly in stressed markets, dealers are less willing to provide capital and customers are instead increasingly stepping up to provide liquidity to other buy-side investors, as demonstrated by O'Hara and Zhou (2021a). Zitzewitz (2010) argues that the simple pairing of orders is a symptom of clients' inability to search over the entire market as limit orders from other investors are typically not visible in OTC markets.

Another trend in U.S. corporate bond trading that is transforming the structure of bond markets and receiving increasing research attention is the shift to more electronic trading. Hendershott and Madhavan (2015) and O'Hara and Zhou (2021b) study electronic RFQ trading and conclude that such electronic trading venues are gaining importance and have substantial implications for liquidity and transaction costs. A more recent development in electronic trading is the emergence of all-to-all trading in the OTC market for corporate bonds, which is impacting the way dealers and buy-side investors interact. Hendershott, Livdan, and Schürhoff (2021) find that quotes from investors in an all-to-all trading environment (i.e., where all market participants can provide liquidity) increases competition, improves prices, and lowers trading costs for investors meaningfully. Another type of electronic trading platform in the U.S. bond market being studied in academic research is alternative trading systems (ATSS). ATSS are trading venues that aggregate and match buyers and sellers to settle transactions. In contrast to exchanges, ATS platforms are private (i.e., available to chosen subscribers only) and thus face fewer regulatory requirements compared

with public exchanges. A subset of ATS platforms called electronic communication networks (ECNs) are organized around a CLOB, much like a public exchange, and offer pre-trade transparency to their venue members. Kozora et al. (2020) find that ATS platforms are preferred when the risk of information leakage is a lesser concern (i.e., for smaller trades in investment-grade corporate bonds). Such centralized platforms are associated with a reduction of 24–32 basis points (bps) in customer transaction costs. Craig et al. (2020) study dealer quotations on two ATS platforms and find that 35% of TRACE-eligible public bonds offer pre-trade transparency on both the bid and ask side. However, such quotations are not publicly disseminated and only available to ATS subscribers, which excludes institutional and retail investors with no access to a certain platform. Overall, the share of electronic trading in corporate bond markets—whether in RFQ trading, all-to-all settings, or ATS venues—is still limited. Moreover, the entrance to electronic venues is often restricted to larger (institutional) investors, while smaller (retail) investors lack access.

Despite these changes in the trading behavior of market participants and evolutions in the market structure over the past decade, the costs for trading corporate bonds appear to remain high and liquidity low. This raises the question of how the current market design, where dealers typically trade bilaterally with clients in opaque trading settings, would benefit from more public transparency. An increasing number of academic papers advocate a market design with public order display facilities that increases the level of information to all market participants prior to trading. Anand et al. (2021) argue that the economic importance of investors supplying direct liquidity through appropriate electronic platforms and trading protocols cannot be overstated. Furthermore, Harris et al. (2015) argue that pre-trade price transparency would increase the market power of less well-informed investors, leveling the bargaining power between clients and dealers. The effect of public pre-trade transparency in corporate bond markets has long been studied only to a limited extent; that is, much of the research has focused on the impact of the post-trade transparency requirements introduced by the TRACE reporting system in 2002.

This study contributes to the increasing literature that assesses price, liquidity, and welfare effects when moving U.S. corporate bond trading from an opaque OTC-driven market design to a more centralized and pre-trade transparent one. By using a proprietary data sample, we are the first to formally compare market metrics of a decentralized, OTC-driven market (i.e., the U.S. corporate bond market) with a bond market where a centralized exchange market coexists with an OTC market (i.e., the Swiss corporate bond market). Such a cross-market comparison is critical as it demonstrates empirically the market quality benefits of a pre-trade transparent market design implemented in practice. Moreover, the Swiss market design includes a duty for dealers to place smaller orders in a public CLOB. Such a market structure, namely a public order display facility combined with an obligation for dealers to route limit orders to such venues, is advocated by researchers such as Harris et al. (2015). The listing of corporate bonds at a central exchange—as is the case in Switzerland—facilitates pre-trade transparency, which can contribute to price discovery. Harris et al. (2015) claim that the U.S. Securities and Exchange Commission (SEC) could quickly and substantially improve bond market efficiency by requiring dealers to post their clients' limit orders to a public CLOB, where orders can be traded among cli-

ents without dealer intermediation. We are the first to study these demanded market structure changes, not through a counterfactual analysis but rather in a market where this market design is already implemented. The Swiss bond market serves as a suitable laboratory where smaller trades are executed in a CLOB trading environment with a high level of pre-trade transparency, while institutional orders are traded OTC between investors and dealers. We demonstrate that such a market design is substantially more beneficial to investors as it entails lower transaction costs, less price discrimination, higher competitiveness, and less impact of orders on prices compared with the OTC-driven U.S. market.

Hong and Warga (2000) are among the first to study the effect of an exchange listing of U.S. corporate bonds on the New York Stock Exchange (NYSE). They find that effective spreads for NYSE-traded corporate bonds tend to be similar to those for dealer-market transactions, despite being at least 50 times smaller than typical orders in the institutional market segment. This is remarkable in that retail orders normally incur substantially higher transaction costs compared with institutional trades due to differences in bargaining power. More recently, Chen and Zhong (2017) investigate pre-trade transparent U.S. corporate bonds listed on the NYSE and study their impact on purely OTC-traded bonds. NYSE-listed corporate bonds see a smaller standard deviation in bid-ask spreads and institutional investors face smaller bid-ask spreads when trading pre-trade transparent bonds. This suggests that pre-trade transparency tends to favor customers rather than dealers by enhancing customers' bargaining capability. They estimate that if NYSE pre-trade transparency was offered as part of OTC trading in U.S. corporate bonds, investors would save \$846 million per year on transaction costs. Holden et al. (2018) also study NYSE-listed corporate bonds and find that they have a 45.8% information share, despite having a tiny market share, while OTC corporate bonds have only a 5.7% information share. They argue that this is because of publicly displayed and frequently updated bid-ask quotes that can be traded at any time. Their results imply that liquidity suppliers are not just passively updating their quotes for NYSE-listed bonds but also actively incorporating new information into their price quotations; thus, they offer market participants a reliable and crucial source of information prior to trading.

Abudy and Wohl (2018) use data from the Tel Aviv Stock Exchange (TASE), where corporate bonds are traded almost exclusively using a CLOB mechanism. They find that, compared with the U.S. market, the small Israeli bond market is more liquid with narrower spreads and lower price dispersion. They argue that these differences are due to the trading mechanism implemented in Israel, where orders are executed in a CLOB. However, the authors are unable to formally compare the two markets, nor can they extend the comparison with OTC trades as in the present paper, simply because off-order-book trading is insignificant in Israel. Odegaard (2017) examines corporate bond trading on the Oslo Stock Exchange (OSE) and finds that trading costs are lower than those for equities, which is in contrast to the evidence from the U.S. where the costs of corporate bonds are much higher than for equities.¹ Roure

¹ For instance, Edwards et al. (2007) estimate average spreads in U.S. equities for retail investors of less than 40 basis points in contrast to 124 basis points estimated for comparable corporate bond trades. Harris and Piwowar (2006) find similar results for the municipal bond market.

et al. (2025) study the interdealer market for German government bonds, where an exchange coexists with bilateral and brokered OTC trading. They document a pecking order of transaction costs: exchange trading is more expensive than OTC, and broker-intermediated trades are more costly than bilateral OTC trades. The exchange provides immediacy and transparency and serves as an outside option for dealers, whereas OTC trading involves search and bargaining frictions. Bilateral OTC trades offer the lowest transaction costs but require counterparties to reveal their identities, while brokered trades provide anonymity at higher cost. The authors highlight that these three protocols play complementary roles and show that dealers' choice of trading protocol is driven by both information and search frictions. Allen and Wittwer (2023) investigate whether to centralize OTC markets, addressing this question in the context of the Canadian government bond market. They document a price gap between institutional investors, who have access to a centralized platform, and retail investors, who do not. They show that many investors do not use the platform despite better prices due to platform costs, weak competition, and dealer-investor relationships. Market centralization can decrease welfare, unless competition is sufficiently strong.

A body of theoretical literature also sheds light on the differences between a market design with a central exchange trading protocol providing a high level of pre-trade transparency and an OTC market with high opacity. Plante (2017) estimates that the transaction costs of U.S. corporate bonds would decrease by 70% if trading migrated from an OTC to a fully centralized CLOB market. In a theoretical model, he estimates that a centralized market structure would be optimal for 91% of the bonds studied, generating a social surplus equal to 1.28% of the total par value for an average bond. In contrast to Plante (2017), who models a full shift to a centralized market, the Swiss bond market is characterized by a dualistic market structure where OTC trading coexists with a centralized exchange. Glode and Opp (2020) compare the coexistence of OTC and limit order markets and argue that OTC trading encourages the acquisition of expertise. Higher expertise in OTC markets can improve allocative efficiency but is problematic if expertise primarily causes adverse selection. In this case, limit order markets tend to dominate over OTC trading. Lee and Wang (2025) develop a model of adverse selection in which dealers price discriminate across customers, offering better bid-ask prices to likely-uninformed customers and thereby "cream skim" them from exchanges into OTC trading. They show that, under certain conditions—particularly when adverse selection risk is low—restricting or even closing OTC trading can improve welfare. Duffie et al. (2017) provide a theoretical model of the effectiveness of benchmarks in opaque OTC markets and study the role of benchmarks to improve market transparency and lower information asymmetries. They conclude that benchmarks in OTC trading can increase social surplus by increasing the volume of beneficial trading, enabling more efficient matching between dealers and investors as well as reducing search costs.

Noteworthy, the U.S. stock markets were organized similarly to today's U.S. corporate bond markets before the SEC adjusted the market design by introducing new order-handling rules on Nasdaq. Given this background, the current structure of the U.S. corporate bond market could be reminiscent of the organization of the Nasdaq stock market prior to the 1997 market reforms. The policy change at the time targeted

Nasdaq's fragmented dealer market and required dealers to display customer limit orders in the order book. Weston (2000) finds that following the implementation of the reforms, spreads in the stock market decreased and trading volume increased—both by 30%. Correspondingly, in 2002, the NYSE introduced OpenBook service, which expanded market participants' access to limit order book information. Boehmer et al. (2005) demonstrate that the liquidity of NYSE stocks increased significantly in that the price impact of orders declined. They argue that pre-trade transparency affects investors' trading strategies and improves liquidity metrics.

Overall, our paper provides direct evidence of a pre-trade transparent trading environment on market quality metrics through a cross-market comparison. Our findings contribute to the current debate on improving liquidity by enhancing pre-trade price information in U.S. corporate bond markets. To the best of our knowledge, our study is the first to formally analyze a market design that is closely aligned with a structure often advocated by researchers and regulators, which has the following three features: (1) a centralized public order display facility that provides pre-trade transparency; (2) a duty for dealers to route limit orders from retail investors to this electronically accessible trading venue; and (3) a coexisting OTC market where institutional investors trade larger orders in opaque trading settings with dealers.

2.2 Economic mechanisms and hypotheses development

The literature on corporate bond market structure suggests that transparency plays a central role in determining market quality. While the introduction of TRACE substantially improved post-trade transparency in the U.S. corporate bond market, pre-trade information remains largely opaque as executable quotes are generally exchanged only between negotiating parties. Consequently, investors possess limited information about prevailing market prices before entering negotiations with dealers.

Several studies suggest that greater pre-trade transparency improves market quality by increasing the availability of publicly observable price information before transactions take place. Harris et al. (2015) advocate a centralized public order display facility that makes executable quotes available to all market participants. Holden et al. (2018) show that publicly displayed quotes on the NYSE contain substantial price information despite the exchange's relatively small market share, while Chen and Zhong (2017) document lower bid-ask spreads and more stable pricing for corporate bonds traded in a pre-trade transparent environment. The common mechanism underlying these studies is that publicly observable executable quotes provide investors with a continuously updated benchmark before trading takes place. A more informative pre-trade environment may reduce information asymmetries, strengthen investors' bargaining position, increase competition among liquidity providers, and ultimately result in lower transaction costs and improved market quality. This reasoning leads to our first hypothesis.

Hypothesis 1 *Corporate bonds traded in a market with publicly observable pre-trade quotes exhibit higher market quality than comparable bonds traded in a predominantly opaque OTC market.*

To examine Hypothesis 1, we compare transaction costs, price competitiveness, and price impact between the Swiss corporate bond market, where corporate bonds are listed in a CLOB, and the predominantly OTC U.S. corporate bond market.

The literature suggests that the benefits of pre-trade transparency are not necessarily confined to transactions executed on the transparent trading venue itself. Harris et al. (2015) argue that publicly displayed limit orders also increase competition in the market. In traditional OTC markets, customer orders are negotiated privately with one or a small number of dealers. By contrast, a CLOB exposes customer trading interest to a broad set of market participants, allowing multiple dealers and other investors to compete for the same order. Publicly displayed executable quotes therefore become transparent reference prices that may also discipline subsequent OTC negotiations, as investors can benchmark dealer quotes against contemporaneous market prices.

Chen and Zhong (2017) provide an explanation of how this mechanism can affect OTC market quality: Rather than improving dealers' ability to identify informed trading, publicly observable executable quotes improve investors' information sets before negotiations begin. Better-informed investors can more effectively evaluate dealer quotes and negotiate from a stronger bargaining position. Dealers consequently face greater competitive pressure and must offer more attractive prices to retain order flow. Thus, they argue that pre-trade transparency reduces transaction costs not because dealers become better informed, but because investors become better informed. This mechanism is relevant particularly for institutional investors: Although large institutional transactions are typically negotiated bilaterally in the OTC market, publicly observable exchange quotes also provide benchmark prices for OTC negotiations. Dealers recognize that institutional investors can compare offered prices with prevailing market conditions, limiting their ability to extract rents.

This mechanism also highlights the distinction between pre-trade and post-trade transparency: Post-trade transparency reveals the prices at which previous transactions were executed, but these prices may no longer reflect current market conditions, particularly in corporate bond markets where trading is infrequent. By contrast, publicly observable executable quotes provide contemporaneous market information before or during negotiations. They therefore may have the potential to influence the bargaining process itself and may improve OTC market quality beyond the informational benefits associated with post-trade transparency alone. Consequently, a CLOB may generate informational spillover effects that benefit market participants even when transactions are ultimately executed in the OTC market.

A related question is whether publicly observable pre-trade quotes provide additional benefits in markets where investors already obtain competing quotes through RFQ protocols. While RFQs allow investors to solicit quotes from multiple dealers, these quotes remain private and are visible only to the participating counterparties. Consequently, investors observe only a limited set of dealer-specific quotes and have little information about whether these quotes accurately reflect prevailing market conditions. By contrast, publicly displayed executable quotes provide a common and continuously observable benchmark before negotiations begin, built from firm dealer quotes as well as potential limit orders from clients. This benchmark may strengthen investors' bargaining position by allowing them to better assess the competitiveness of dealer quotes.

The Swiss corporate bond market provides a unique setting to examine the mechanism of pre-trade transparency because a transparent CLOB coexists with an active OTC market: While many institutional transactions continue to be negotiated bilaterally, market participants can continuously observe executable quotes in a CLOB. If these quotes serve as benchmarks during OTC negotiations, the benefits associated with pre-trade transparency should extend beyond exchange trading and also be reflected in OTC transactions. This reasoning leads to our second hypothesis.

Hypothesis 2 *The improved market quality associated with pre-trade transparency extends beyond CLOB trading and is also reflected in OTC transactions.*

To test Hypothesis 2, we compare the effective bid-ask spreads of OTC transactions in the Swiss and the U.S. corporate bond markets. In addition, we compare effective bid-ask spreads for transactions executed through a CLOB trading mechanism in both markets to assess whether the observed differences are specific to OTC trading.

Directly identifying the causal effect of publicly observable quotes on OTC negotiations is inherently difficult. However, if CLOB quotes serve as reference prices during bilateral negotiations, OTC transaction prices should generally be consistent with quoted bid-ask prices. Examining the relationship between CLOB quotes and OTC transaction prices therefore provides additional evidence on the proposed bargaining mechanism. This reasoning leads to our third hypothesis.

Hypothesis 3 *OTC transaction prices in the Swiss corporate bond market remain largely within the average daily bid-ask quotes displayed on the CLOB.*

Sections 4 and 5 examine these hypotheses. Section 4 analyzes whether pre-trade transparency is associated with higher overall market quality, while Section 5 investigates whether these benefits extend to OTC transactions and whether CLOB quotes provide meaningful benchmarks for OTC pricing.

3 Market description and data

In this section, Chapters 3.1 and 3.2 describe the structure and functioning of the two corporate bond markets and identify the main differences. Chapter 3.3 describes our data sets that are used for the subsequent analyses.

3.1 The Swiss corporate bond market

The Swiss bond market is characterized by a centralized market structure with an exchange at its core and an OTC market existing alongside it. The SIX Swiss Exchange (hereinafter “SIX”) serves as the primary exchange² in Switzerland and

² There is another exchange in Switzerland, BX Swiss. However, the incidence of CHF-denominated bond listings is negligible.

is owned by a broad consortium of domestic and international financial institutions. Members of the exchange are authorized to submit quotes and orders for listed securities, either on behalf of clients or for proprietary trading. Debt securities admitted to trading on SIX are issued by both domestic and foreign entities and encompass a wide range of instruments, such as straight bonds, floating-rate notes, and asset-backed securities in all major currencies. The issuers include governments, sub-sovereigns, supranationals, agencies, and corporations. SIX operates the following four bond market segments: “Bonds—CHF Swiss Confederation”,³ “Bonds—CHF Swiss Pfandbriefe”,⁴ “Bonds—CHF”, and “Bonds—Non CHF”. In this paper, we focus on the “Bonds—CHF” segment, namely Swiss franc-denominated bonds issued by domestic or foreign companies. We exclude bonds issued by sub-sovereigns, multinationals, and agencies and focus on corporate bonds (hereinafter “CHF-bonds”).

The market structure in CHF-bond trading is characterized by two distinct features: First, nearly all CHF-bonds issued are exchange-listed on SIX. Although no legal obligation exists for an issuer to list CHF-bonds on the exchange, most companies follow the informal market standard and voluntarily list CHF-bonds on SIX. The reasons for an exchange listing include improved visibility to investors; greater investor confidence, as issuers are subject to certain transparency requirements; and better tradability for investors. Moreover, the Swiss Bond Index (SBI), which serves as a benchmark for the Swiss franc capital market, requires CHF-bonds to be listed on SIX. As certain institutional investors in Switzerland rely on the SBI’s inclusion criteria when defining their investment universe, bond issuers have an incentive to list issues on the exchange to broaden their investor base. Second, until February 1, 2019, trading members had a duty to trade orders with a nominal value smaller than CHF 100,000 on the exchange.⁵ This means that dealers (i.e., banks and brokers) are required to post smaller client orders on the exchange and are not allowed to execute retail orders OTC (i.e., off the order book). As a result, OTC trading is essentially nonexistent in this trade size category, and smaller orders are typically executed on the order book.

The trading model for CHF-bonds on SIX is continuous CLOB trading, which allows for limit quotes and iceberg orders. Orders from customers and quotes from dealers acting as liquidity providers compete against each other in the CLOB. Trading is based on orders and quotes, while information on price and volume is available to all market participants. The public dissemination of price information provides pre-trade transparency to all market participants. SIX charges its members a transaction fee of CHF 1.5 plus a value fee ranging from 0 for liquidity providers to 0.01% for client orders. The value fee is floored and capped at CHF 2 and 100, respectively. These costs are not reflected in our data sample.

³ This segment comprises Swiss government bonds, known as “Eidgenossen”.

⁴ “Pfandbriefe” are similar to covered bonds in other markets. In Switzerland, Pfandbriefe are issued by two institutions to help banks raise funds more efficiently.

⁵ The obligation to trade on the exchange has been lifted on February 1, 2019, i.e., after our analysis which refers to 2018 when the duty to trade on the exchange was still in place. The impact of removing these order-handling requirements for dealers provides interesting scope for future research.

Beside trading on a central exchange, there coexists an OTC market in which CHF-bonds are traded bilaterally between dealers and customers. Liquidity providers that quote prices on the exchange typically also offer price quotes on an RFQ basis in the OTC market. During trading hours, orders executed OTC must be reported to the exchange within 15 min. SIX publishes the time, price, and volume of an OTC trade immediately after reporting it in the trade book. Transactions executed outside of the continuous trading hours are published before opening on the next trading day. At the request of market participants, larger transactions or trades in illiquid bonds may be published with a delay until the opening of the next trading day.

At the beginning of 2018, the outstanding volume of CHF-denominated corporate bonds amounted to CHF 207 billion. Compared with the U.S., where the issuance volume is \$10,590 billion, the Swiss corporate bond market is approximately 50 times smaller.⁶

3.2 The U.S. corporate bond market

The structure of the U.S. corporate bond market differs from that of Switzerland in that it is more decentralized, with transactions predominantly negotiated OTC between dealers and clients. As securities are typically not listed on a centralized exchange, only limited pre-trade transparency is available to the investing public. Instead, dealers provide price quotes to customers on a bilateral RFQ basis, which limits the availability of pre-trade price information to other market participants not involved in the transaction. Although the U.S. bond market has become more transparent post-trade following the introduction of the TRACE reporting system in 2002, investors still have limited access to pre-trade price information, as dealer quotations are not publicly disseminated. Chen and Zhong (2017) argue that some potential data sources provide quote data, such as proprietary market information vendors (e.g., Bloomberg), electronic trading platforms (e.g., MarketAxess and Tradeweb), and private trading networks (e.g., BlackRock's Aladdin Trading Network); however, these data sources are fragmented, not publicly accessible, and typically offer only indicative rather than live executable price quotes, making their use as pre-trade price information less reliable for investors. This represents a key difference in market structure relative to Switzerland, where bonds are exchange-listed, providing investors with a high degree of pre-trade transparency.

Noteworthy, the U.S. corporate bond market has historically been organized similarly to the Swiss market, offering a high degree of pre-trade transparency through centralized exchange listings. Biais and Green (2019) find that in the earlier 1920s, the majority of corporate bonds were listed and traded on the NYSE's transparent CLOB. By the mid-1940s, activity declined significantly as trading shifted from the NYSE to the OTC market. The authors argue that this shift in the market structure coincided with a remarkable growth in the weight and importance of institutional investors (i.e., insurance companies and pension funds). They claim that institutional investors were better off in an OTC environment, where they could effectively exploit their bargaining power, without being hindered by reporting and price priority constraints. Today,

⁶ Both figures refer to our cleaned data samples.

only a minority of bonds are listed on the NYSE Bonds market offering (previously known as the Automated Bond System; ABS). The NYSE Bonds market is arguably the most relevant public exchange in the U.S. for corporate bonds, offering a CLOB system with a transparent trading environment of live and executable price quotes.⁷ However, Edwards et al. (2007) find that only approximately 5% of all bonds traded between 2003 and 2005 are listed on the NYSE. In a more recent data sample, Chen and Zhong (2017) find that only approximately 12% of the securities traded between 2005 and 2015 are listed on the NYSE, which confirms the perception that most corporate bonds are traded only in the OTC market. Unlike OTC trades, transactions executed on a public U.S. securities exchange are not required to be reported to the Financial Industry Regulatory Authority's (FINRA) TRACE database, given that transaction information is disseminated publicly by the exchange. In its "Report to Congress on Capital Access and Market Liquidity", the U.S. Securities and Exchange Commission (2017) notes that without a central exchange, the corporate bond market is highly fragmented and lacks broadly disseminated pre-trade pricing information.

Instead of public exchanges, ATSs are a more established form of centralized trading in the U.S. corporate bond market. Similar to a CLOB offered by exchanges, ATSs may also use order protocols that aggregate and match orders. Kozora et al. (2020) argue that ATS platforms either use an RFQ protocol, where platforms solicit bids and offers electronically, which counterparties can meet, or they use automated matching systems, which provide immediately executable liquidity by aggregating orders from market participants, similar to a CLOB. Kozora et al. (2020) refer to the latter method as electronic communication networks (ECNs) and categorize such electronic limit order books as a prominent subset of ATS platforms that allow market participants to receive an immediate execution. They find that the majority of trades executed on ATS platforms are pursuant to an ECN protocol that involves a live and executable limit order book. The BIS Markets Committee (2016) argues along the same lines, noting that ECNs often employ a CLOB trading mechanism in which transactions are based on orders and quotes, comparable to a CLOB offered by public exchanges, such as NYSE Bonds. Kozora et al. (2020) find that in 2017, the monthly trading volume on ATS platforms represented 2.1% of the total corporate bond market volume and 16.1% of all trades, suggesting that ATS platforms primarily facilitate smaller transactions. Although ATS platforms offer a certain level of pre-trade transparency, Craig et al. (2020) argue that live quotations on ATSs are generally only available to ATS members or through quote consolidators to a limited group of participants. As a result, the U.S. corporate bond market continues to exhibit relatively limited pre-trade transparency compared with the Swiss market, where price information is publicly available through an exchange-based trading system.

⁷ Contrary to SIX in Switzerland, the NYSE implemented several obligations for designated liquidity providers to maintain liquidity in the listed bonds. For instance, they are required to maintain a bid or offer at the exchange's best bid or ask for at least 5% of the trading day for assigned bonds in the aggregate.

3.3 Data samples

This paper uses two data samples of corporate bond transactions to study cross-market comparisons between the U.S. and Swiss markets. For the Swiss market, we use a proprietary data sample provided by SIX that covers transactions in CHF-bonds. The sample includes all transactions in CHF-bonds that were executed in the order book on SIX or traded bilaterally in the OTC market. We consider only corporate bonds and exclude covered bonds or securities issued by sub-sovereigns, supranationals, and agencies. The transaction sample of CHF-bonds consists of trade-level information and includes the date and time, the price, and the volume of a transaction. The data sample identifies the counterparties involved in a transaction, allowing the classification of trades as dealer-to-client or inter-dealer transactions. However, the data sample does not include the direction of a trade. In addition to the trade-level information, the data contain quoted data in all CHF-bonds from the order book of SIX. This covers the daily average bid-ask prices and the average daily volume quoted on the exchange. Beside credit ratings from S&P and Moody's, we use credit ratings from Zurich Cantonal Bank (ZKB). Knoblauch (2021) finds that only approximately half of the CHF-bonds in Switzerland are rated by the two international rating agencies, especially larger issues from foreign corporations. Smaller issues from domestic borrowers are often rated by local banks such as ZKB. We refer to the data sample with CHF-bond transactions as the "SIX-sample".

For the U.S. corporate bond market, we retrieve transactions from the enhanced version of FINRA's TRACE database. Dealers are required to report OTC secondary-market transactions in U.S. corporate bonds to TRACE. The data consist of trade-level information, such as the date and time of a transaction, price, and uncapped information on the order volume. Credit ratings are used from S&P and Moody's. Similar to the SIX-sample, TRACE recognizes the counterparties involved in a transaction. It also identifies trades executed electronically on an ATS platform. As ATSs run different trading protocols, we are unable to determine the trading mechanism through which a trade was executed. However, Kozora et al. (2020) argue that the majority of trading on the most active ATS platforms occurs through a trading protocol that involves a live and executable order book. The BIS Markets Committee (2016) supports this claim, arguing that ATS transactions are often traded pursuant to a CLOB trading mechanism where prices and volumes are transparent pre-trade. We therefore interpret ATS-flagged trades in the TRACE database as being executed through protocols broadly comparable to those used for CHF-denominated bonds traded in the CLOB of SIX. We refer to this dataset of U.S. corporate bond transactions as the "TRACE sample". Both datasets cover the full year 2018.

4 Differences in liquidity measures

In this section, Chapter 4.1 initially presents the summary statistics for the two data samples, before examining the liquidity measures for the two markets separately by trading activities and transaction prices in Chapter 4.2 and 4.3, respectively.

4.1 Summary statistics

Tables 1 and 2 report summary statistics for the two data samples and the two corporate bond markets. In addition to the full market sample, the tables present summary statistics for bonds traded in the OTC market and on electronic trading venues (CLOB from SIX in Switzerland and ATS in the U.S.). The average USD/CHF exchange rate was around parity in 2018, allowing for a direct comparison of values across currencies.

The two markets differ substantially in size. The U.S. sample includes 17,024 bonds and approximately 14.9 million trades, whereas the CHF-denominated sample comprises 1,070 bonds and 267,009 trades. This difference is also reflected in trading activity: the U.S. market averages about 60,000 daily trades, compared with approximately 1,000 in the Swiss market. Transaction size distributions also differ across markets: The median trade size is higher in Switzerland (35,000 vs. 25,000), whereas the mean trade size is substantially larger in the U.S. market (414,737 vs. 212,017). This pattern suggests that the U.S. market is characterized by a greater prevalence of large block trades, which inflate average transaction sizes. Consistent with this interpretation, the maximum trade size reaches 856 million in the U.S. market, compared with 100 million in CHF-denominated bonds. Differences in market size are also reflected in bond issuance sizes: The median issuance volume of U.S. corporate bonds (500 million) exceeds that of CHF-denominated bonds (200 million). In addition, the largest bond issues in the U.S. market (15 billion) are approximately ten times larger than those in Switzerland (1.65 billion), suggesting a greater prevalence of large “jumbo” issuances.

Bond characteristics are broadly similar across the two markets. Median bond age is 3.4 years in Switzerland and 3.7 years in the U.S., while the median remaining maturity is 4.5 and 4.9 years, respectively. Mean values are substantially higher in both markets, indicating that the distribution is skewed by a small number of bonds with very long maturities and high ages. However, mean values are higher in the U.S. market, indicating a longer right tail of older and longer-maturity bonds.

The two markets differ more noticeably with respect to credit quality: The median Swiss bond is rated A, whereas the median U.S. bond is rated BBB+, two notches lower. In addition, the U.S. corporate bond market has a higher proportion of high-yield bonds: 20% of U.S. bonds are rated below BBB, compared with only 3% of CHF-denominated bonds. This indicates that bonds in the Swiss market are, on average, of higher credit quality. At the same time, the share of unrated bonds in the CHF-bond sample is substantially higher. Knoblauch (2021) notes that larger CHF-bonds of foreign issuers with higher ratings are usually rated by the international rating agencies S&P, Moody's, or Fitch, while domestic issuers with lower ratings (below A-) are often covered by local banks or rating providers (e.g., ZKB, UBS, or Fedafin). To account for this, we also incorporate credit ratings from Zurich Cantonal Bank (ZKB) for CHF-denominated bonds when ratings from S&P or Moody's are not available. However, some unrated CHF-bonds in our sample may still correspond to lower-rated securities for which rating information is unavailable.

Tables 1 and 2 further report descriptive statistics separately for bonds traded in the OTC market and on electronic trading venues. In both markets, bonds that trade

Table 1 Summary statistics Swiss corporate bond market

	Median	Mean	Min	Max
<i>Market view: n=1,070 bonds (267,009 trades)</i>				
Daily volume (in CHF millions)	220	227	87	679
Number of daily transactions	1,059	1,073	624	1,550
Transaction size	35,000	212,017	448	100,000,000
Issue size (in CHF millions)	200	228	25	1,650
Age	3.4	4.0	0.0	33.3
Maturity	4.5	5.1	0.0	34.3
Rating	A	A	B–	AAA
Investment-grade	77%			
High-yield	3%			
Not rated	20%			
<i>OTC channel: n=1,048 bonds (40,026 trades)</i>				
Daily volume (in CHF millions)	170	179	51	630
Number of daily transactions	159	161	61	447
Transaction size	500,000	1,113,481	5,000	100,000,000
Issue size (in CHF millions)	200	229	25	1,650
Age	3.4	3.9	0.0	33.0
Maturity	4.5	5.1	0.0	34.2
Rating	A	A	B–	AAA
Investment-grade	78%			
High-yield	3%			
Not rated	19%			
<i>CLOB channel: n=1,055 bonds (226,983 trades)</i>				
Daily volume (in CHF millions)	47	48	24	97
Number of daily transactions	904	912	551	1,369
Transaction size	25,000	53,053	448	5,000,000
Issue size (in CHF millions)	200	228	25	1,650
Age	3.4	3.9	0.0	33.3
Maturity	4.5	5.0	0.0	34.3
Rating	A	A	B–	AAA
Investment-grade	77%			
High-yield	3%			
Not rated	20%			

This table presents descriptive statistics for the Swiss corporate bond sample (the SIX-sample). The upper panel reports statistics for the full market sample of CHF-bonds, while the lower panels report statistics for bonds traded in the OTC market and in the CLOB of SIX, respectively. For each sample, the table reports the median, mean, minimum, and maximum values of daily trading volume, the number of daily transactions, transaction size, issue size, bond age, maturity, and credit rating. The rating distribution shows the proportions of investment-grade, high-yield, and unrated bonds. High-yield bonds are defined as bonds rated below BBB. Credit ratings are used from S&P or Moody's; if unavailable, we use the ratings from Zurich Cantonal Bank (ZKB). Following Abudy and Wohl (2018), bonds with an issue size below CHF 25 million are excluded. The sample includes both dealer-to-client and interdealer trades and covers the year 2018

electronically are characterized by markedly smaller transaction sizes than bonds traded OTC. In Switzerland, the median transaction size is 25,000 in the CLOB of SIX compared with 500,000 in the OTC market. Similarly, in the U.S., ATS trades have a median size of 20,000, whereas the corresponding OTC median amounts to

Table 2 Summary statistics U.S. corporate bond market

	Median	Mean	Min	Max
<i>Market view: n=17,024 bonds (14,869,575 trades)</i>				
Daily volume (in USD millions)	25,588	24,767	2,004	37,293
Number of daily transactions	60,367	59,717	16,101	82,296
Transaction size	25,000	414,737	101	856,162,000
Issue size (in USD millions)	500	723	25	15,000
Age	3.7	5.4	0.0	88.4
Maturity	4.9	8.4	0.0	150.9
Rating	BBB+	BBB	D	AAA
Investment-grade	74%			
High-yield	20%			
Not rated	6%			
<i>OTC channel: n=17,018 bonds (10,715,400 trades)</i>				
Daily volume (in USD millions)	23,865	23,212	1,757	35,471
Number of daily transactions	43,463	43,034	11,478	60,232
Transaction size	35,000	539,381	101	856,162,000
Issue size (in USD millions)	500	723	25	15,000
Age	3.3	4.9	0.0	88.4
Maturity	4.9	8.4	0.0	150.9
Rating	BBB+	BBB	D	AAA
Investment-grade	74%			
High-yield	20%			
Not rated	6%			
<i>ATS channel: n=13,497 bonds (4,154,175 trades)</i>				
Daily volume (in USD millions)	1,550	1,555	240	2,429
Number of daily transactions	16,708	16,683	4,623	22,739
Transaction size	20,000	93,225	107	37,500,000
Issue size (in USD millions)	500	749	25	15,000
Age	3.7	5.3	0.0	88.4
Maturity	5.1	8.7	0.0	99.9
Rating	BBB+	BBB+	D	AAA
Investment-grade	80%			
High-yield	17%			
Not rated	3%			

This table presents descriptive statistics for the U.S. corporate bond sample (the TRACE-sample). The upper panel reports statistics for the full market sample of TRACE-bonds, while the lower panels report statistics for bonds traded in the OTC market and on ATS, respectively. For each sample, the table reports the median, mean, minimum, and maximum values of daily trading volume, the number of daily transactions, transaction size, issue size, bond age, maturity, and credit rating. The rating distribution shows the proportions of investment-grade, high-yield, and unrated bonds. High-yield bonds are defined as bonds rated below BBB. Credit ratings are used from S&P. Following Abudy and Wohl (2018), bonds with an issue size below USD 25 million are excluded. The sample includes both dealer-to-client and interdealer trades and covers the year 2018

35,000. Moreover, the prevalence of trading through a CLOB mechanism differs markedly across the two markets: In the U.S., approximately 28% of all transactions are executed on an ATS, whereas in the Swiss corporate bond market about 85% of the trades are traded in the CLOB of SIX. Section 5.1 examines these differences in trading channel usage in greater detail.

Based on the univariate comparison of the aggregate bond characteristics, the U.S. corporate bond market would be expected to exhibit higher liquidity than the Swiss market. Prior research shows that bond issuance size is a key determinant of liquidity (e.g., Hotchkiss and Jostova 2017), and U.S. corporate bonds are characterized by substantially larger issuance volumes and significantly higher trading activity. Average daily trading volume amounts to \$24,767 million in the U.S., compared with CHF 227 million in Switzerland. Differences in overall market size further reinforce this expectation: The total outstanding volume of U.S. corporate bonds is approximately \$10,590 billion, compared with CHF 207 billion in Switzerland, making the U.S. market roughly 50 times larger.⁸ Taken together, these factors suggest that the U.S. market should, *ex ante*, be more liquid.

We next examine whether these expectations are borne out in the data. Following Hameed et al. (2019), we distinguish between quantity-based and price-based liquidity measures, which we analyze in Sects. 4.2 and 4.3, respectively.

4.2 Quantity-based measures

Table 3 presents quantity-based measures of liquidity in the two markets. These measures capture trading intensity, with higher values indicating greater liquidity. The turnover rate is defined as monthly trading volume relative to the amount outstanding. The number of trades measures trading activity per bond per month, while the number of trading days indicates the percentage of days within a month on which a bond is traded. All measures are computed at the bond-month level.

Table 3 Quantity-based liquidity measures in the two markets

	Turnover rate (monthly)	Number of trades per bond (monthly)	Number of trading days (% of month)
<i>Switzerland</i>			
95th	8.2%	86	94%
75th	2.8%	26	60%
50th	1.3%	12	35%
25th	0.5%	6	18%
5th	0.1%	2	5%
Mean	2.3%	25	41%
<i>U.S.</i>			
95th	16.3%	338	100%
75th	5.5%	89	90%
50th	2.2%	28	57%
25th	0.5%	6	21%
5th	0.0%	2	5%
Mean	4.5%	80	55%

This table presents quantity-based measures of liquidity for the U.S. and Swiss corporate bond markets. The table presents quantiles for the measures and includes the mean in the last row. The upper table displays CHF-bonds' characteristics in the Swiss market (the SIX-sample), while the lower table reports transactions and bonds in the U.S. corporate bond market (the TRACE-sample). The sample period is 2018. The figures are measured monthly for each bond

⁸ As of the beginning of 2018 and referring to our cleaned data samples from Tables 1 and 2.

The results indicate higher levels of quantity-based liquidity in the U.S. market, consistent with the expectation of a more active trading environment than in CHF-denominated bonds. For example, the median monthly turnover rate is 2.2% in the U.S., compared with 1.3% in Switzerland, implying that a typical bond turns over approximately once every four years in the U.S. market versus nearly seven years in Switzerland. Trading activity shows a similar pattern: The median number of trades per bond per month is 28 in the U.S. and 12 in Switzerland. In addition, the median proportion of trading days is 57% in the U.S. and only 35% in Switzerland, indicating that a typical Swiss bond trades on only about one-third of trading days. Overall, these measures confirm that the Swiss market is not only smaller but also characterized by lower trading activity and liquidity relative to the U.S. market, which itself is often considered illiquid in the literature.

As shown in the previous section, the two samples differ along several bond characteristics. Consequently, the differences in quantity-based liquidity measures reported in Table 3 may partly reflect differences in sample composition. This implies that a direct comparison of U.S. corporate bonds and CHF-denominated bonds may be affected by selection bias. To account for these differences, we control for observable bond characteristics by estimating cross-market differences in liquidity measures, following O'Hara and Zhou (2021a, b). Specifically, we estimate the following model:

$$\text{Measure}_{i,ISize,Age,Maturity,Rating,m} = \alpha + \beta \times \text{Market}_{i,m} + \mu_{ISize,Age,Maturity,Rating,m} + \varepsilon_{i,ISize,Age,Rating,Maturity,m} \quad (1)$$

The dependent variable $\text{Measure}_{i,ISize,Age,Maturity,Rating,m}$ describes the average liquidity measures from Table 3 and refers to bond i with the same issue size ($ISize$), same number of days since issuance (Age), same time to expiry ($Maturity$), and same credit quality ($Rating$), and also traded in the same month m . The variable $\text{Market}_{i,m}$ is a dummy for CHF-bonds. We include the fixed effect variable $\mu_{ISize,Age,Maturity,Rating,m}$, which controls for size, age, maturity, rating, and month fixed effects. This enables us to compare the monthly liquidity measures from Table 3 for U.S. and Swiss bonds that have the same issue size, age, maturity, and credit rating. Standard errors are clustered at the bond and month levels.

Table 4 confirms the qualitative findings from Table 3 and shows that quantity-based liquidity measures are higher in the U.S. corporate bond market than in the Swiss market. For example, the average monthly turnover rate in CHF-denominated bonds is 1.8% points lower, indicating that a larger fraction of outstanding volume is traded each month in the U.S. market. A similar pattern emerges for trading activity: The average number of trades per month is lower in Switzerland than for U.S. corporate bonds with comparable characteristics. Consistent with this, the share of trading days per month is significantly lower in the market for CHF-bonds, implying that bonds are traded more frequently over time in the U.S. market. Overall, these results reinforce the evidence from the summary statistics, indicating that trading intensity is substantially higher in the U.S. corporate bond market than in the Swiss market.

Hameed et al. (2019) argue that markets with lower trading activity can still exhibit high levels of liquidity, for example by maintaining competitive pricing and

Table 4 Estimated differences in quantity-based liquidity measures in the two markets

	Turnover rate	Number of trades	Number of trading days
<i>CHF-bonds</i>	− 0.0181*** (0.0006)	− 26.53*** (0.6223)	− 0.0306*** (0.0032)
Fixed effect:			
<i>Size-age-maturity-rating-month FE</i>	Yes	Yes	Yes
Observations	169,944	169,944	169,944
R2	0.10	0.12	0.21

This table presents the results from estimating Eq. (1). Separate models are run to estimate the quantity-based liquidity measures listed in Table 3. The variable *CHF-bonds* is a market dummy for bonds in the Swiss market (the SIX-sample) and indicates differences in the liquidity measures compared with the U.S. market (the TRACE-sample), thus capturing differences in the market design. *Size-age-maturity-rating-month fixed effect* controls for measures in bonds with the same issue size, age, maturity, and credit rating category, and traded in the same month. The categories for the fixed effect variables are defined according to Hendershott and Madhavan (2015) as follows: *issue size* (small: <100 m, medium: 100–500 m, large: 500+ m), *age* (<2 y, 2–5 y, 5+ y), *maturity* (<2 y, 2–5 y, 5–20 y, 20+ y), *rating* (superior: AA and up, other investment-grade: BBB–A, high-yield: below BBB, not rated). Please refer to Table 3 for a description of the dependent variables. The sample period is 2018. Standard errors are clustered at the bond and month levels

low transaction costs. In this sense, bonds may trade less frequently—as in the Swiss market—yet remain liquid if transactions have only a limited impact on prices. Thus, high trading activity or volume alone is not sufficient to assess market quality. Motivated by this distinction, we next examine price-based measures of liquidity across the two markets.

4.3 Price-based measures

Price-based liquidity measures capture the cost dimension of liquidity by examining impact of trades on prices, price competitiveness, and transaction costs. Table 5 reports summary statistics for three such measures, based on dealer-to-client transactions in the U.S. and Swiss corporate bond markets. All measures are calculated at the bond-level.

First, price dispersion captures the competitiveness of prices in a market. Randall (2015) argues that higher price dispersion is associated with lower price competitiveness. Following Abudy and Wohl (2018), we measure price dispersion as the daily standard deviation of transaction prices relative to the average daily price of a bond. The measure is calculated at the bond-level. Table 5 shows that price dispersion is substantially higher in the U.S. corporate bond market than in Switzerland. The median price dispersion is 0.25% in the U.S., compared with only 0.06% in the Swiss market. This pattern is consistent with less competitive pricing in the U.S. market and a higher degree of price dispersion across transactions. However, differences in price dispersion may also partly reflect differences in the number of observed transactions per bond.

Second, the Amihud (2002) illiquidity ratio measures the price impact of trades. Following Dick-Nielsen et al. (2012), we calculate the Amihud measure as the price response associated with one million units traded and exclude retail transactions in order to focus on institutional trading activity. For each bond, the measure is computed as the daily average of the absolute return between consecutive transactions

Table 5 Price-based liquidity measures in the two markets

	Price dispersion	Amihud illiquidity ratio	Effective bid-ask spread
<i>Switzerland</i>			
95th	0.21%	0.86%	0.59%
75th	0.10%	0.26%	0.25%
50th	0.06%	0.10%	0.16%
25th	0.03%	0.02%	0.09%
5th	0.00%	0.00%	0.01%
mean	0.09%	0.22%	0.23%
<i>U.S</i>			
95th	1.01%	2.57%	1.66%
75th	0.44%	0.57%	0.71%
50th	0.25%	0.27%	0.38%
25th	0.13%	0.11%	0.21%
5th	0.03%	0.01%	0.06%
mean	0.35%	0.70%	0.56%

This table presents price-based liquidity measures of the U.S. and Swiss corporate bond markets. Price dispersion and the Amihud illiquidity ratio are first calculated at the bond-day level and subsequently aggregated to the bond-month level. Effective bid-ask spreads are first calculated for individual dealer-to-client transactions, averaged within each bond-day, and then aggregated to the bond-month level. The table presents quantiles for these monthly measures per bond. The upper table displays CHF-bonds' characteristics in the Swiss market (the SIX-sample), while the lower table reports transactions and bonds in the U.S. corporate bond market (the TRACE-sample). Price dispersion measures the price competitiveness and is defined as the daily standard deviation of prices as a percentage of the average price, following Abudy and Wohl (2018). The Amihud (2002) illiquidity ratio measures the price impact of a trade per unit traded (in millions). Following Dick-Nielsen et al. (2012), retail trades are excluded. The effective bid-ask spread measures transaction costs and is calculated similarly to Choi et al. (2024), using the daily weighted-average price of all interdealer trades as the reference price. Since the SIX-sample does not identify information on the trade direction, we take the absolute value of the relative costs from a trade. The table regards only dealer-to-client trades. The sample period is 2018

divided by the corresponding trade size. At least two transactions are required to compute the daily measure. The monthly Amihud measure is then defined as the median of the daily values within a month. The measure is calculated at the bond-level. Table 5 shows that the median price impact of institutional trades is lower in the Swiss market than in the U.S. market. A dealer-to-client trade of one million moves prices by 0.10% in Switzerland, compared with 0.27% in the U.S. market. This suggests that the Swiss market is able to absorb trades of a given size with a smaller impact on prices.

Third, the effective bid-ask spread measures transaction costs. Following Choi et al. (2024), we compute effective spreads relative to the volume-weighted average price of interdealer trades within a given day. Daily effective spreads are averaged across dealer-to-client transactions, monthly values are obtained as the median of the daily averages, and the final measure is calculated at the bond level as the median across all monthly observations. Table 5 shows that transaction costs are substantially higher in the U.S. corporate bond market than in Switzerland. The median effective bid-ask spread is 0.38% in the U.S. market, compared with only 0.16% for CHF-denominated bonds. This finding is particularly striking given the substantially larger size and higher trading activity of the U.S. market.

Figure 1 illustrates the evolution of the three price-based liquidity measures across the two markets over time. The results show that the cost dimension of liquidity is consistently more favorable in the Swiss market throughout the 2018 sample period. Despite substantially higher trading activity in the U.S. corporate bond market, Switzerland exhibits lower price dispersion, lower price impact, and lower transaction costs.

Similar to the analysis of quantity-based liquidity measures, the observed differences in price-based liquidity may partly reflect differences in sample composition. To account for this, we follow O'Hara and Zhou (2021a) and estimate the measures from Table 5 using an extended version of Eq. (1). In addition to the *size-age-maturity-rating-month* fixed effects used previously, we estimate separate specifications that include additional categorical controls shown in the literature to be important determinants of price-based liquidity.

$$Measure_{i,m} = \alpha + \beta_1 \times Market_{i,m} + X_{ISize, Age, Maturity, Rating} + \mu_m + \epsilon_{i,m} \quad (2)$$

The variable $Market_{i,m}$ is a dummy variable indicating CHF-bonds. The categorical control variables $X_{ISize, Age, Maturity, Rating}$ are constructed following Hendershott and Madhavan (2015), and their levels are described in Table 4. Table 6 reports the estimated cross-market differences in the price-based liquidity measures.

Table 6 confirms the qualitative findings from Table 5 and Figure 1 and shows that price-based liquidity measures are more favorable in the Swiss corporate bond market than in the U.S. market. For example, column (6) indicates that effective bid-ask spreads are, on average, 0.37% points lower for CHF-denominated bonds, implying substantially lower transaction costs for investors. Similarly, column (2) shows that price dispersion is 0.28% points lower in the Swiss market, consistent with a higher degree of price competitiveness. In addition, the price impact of trades is significantly smaller for CHF-bonds than for comparable U.S. corporate bonds. Taken together, these results suggest that despite its substantially smaller size, the

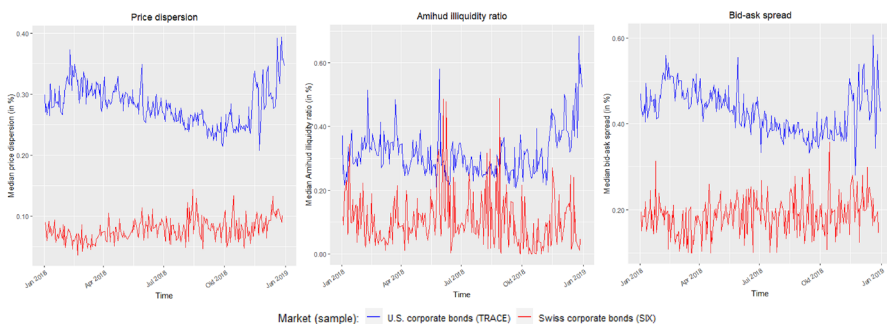


Fig. 1 Time series of price-based liquidity measures. The panels in this figure present time series of the price-based liquidity measures from Table 5. For the time series plots, the measures are calculated for each bond on each date; then, we take the median over all bonds per day. All measures are calculated at the bond-level. The red lines indicate median measures for the Swiss market (the SIX-sample), while the blue lines indicate them for the U.S. market (the TRACE-sample). Please refer to Table 5 for a description of the variables. The sample period is 2018

Table 6 Estimated differences in price-based liquidity measures in the two markets

	Price dispersion		Amihud illiquidity ratio		Effective bid-ask spread	
	(1)	(2)	(3)	(4)	(5)	(6)
CHF-bonds	- 0.3105*** (0.0039)	- 0.2798*** (0.0039)	- 0.6110*** (0.0255)	- 0.4978*** (0.0295)	- 0.3683*** (0.0088)	- 0.3701*** (0.0096)
Issue si=e (100-500 m)	- 0.3423*** (0.0121)		- 2.058*** (0.1929)		- 0.4527*** (0.0147)	
Issue si=e (500+m)	- 0.4205*** (0.0122)		- 2.272*** (0.1947)		- 0.5244*** (0.0146)	
Age(2-5y)	0.0675*** (0.0024)		0.1981*** (0.0128)		0.0484*** (0.0042)	
Age(5+y)	0.1801*** (0.0030)		0.6057*** (0.0185)		0.1547*** (0.0046)	
Maturity (2-5 y)	0.1052*** (0.0025)		0.1697*** (0.0148)		0.1669*** (0.0043)	
Maturity (5-20 y)	0.2761*** (0.0028)		0.5305*** (0.0167)		0.4406*** (0.0044)	
Maturity (20+y)	0.3502*** (0.0042)		0.6613*** (0.0202)		0.6253*** (0.0073)	
Rating (BBB-A)	0.0359*** (0.0026)		0.0732*** (0.0128)		0.0719*** (0.0051)	
Rating (<BBB)	0.2112*** (0.0041)		0.5678*** (0.0223)		0.4201*** (0.0073)	
Rating (unrated)	0.2415*** (0.0096)		0.1212** (0.0398)		0.5732*** (0.0180)	
<i>Fixed effect:</i>						
Month	Yes	No	Yes	No	Yes	No
rating-month FE	No	Yes	No	Yes	No	Yes
Observations	104,984	104,984	107,553	107,553	137,192	137,192
R2	0.23	0.29	0.06	0.12	0.16	0.21

This table presents the results from the regression models displayed in Eqs. (1) and (2) to evaluate the determinants of price-based liquidity measures listed in Table 5. The variable *CHF-bonds* is a market dummy for bonds in the Swiss market (the SIX-sample) and indicates differences in liquidity measures compared with the U.S. market (the TRACE-sample), thus capturing differences in the market design. *Size-age-maturity-rating-month fixed effect* controls for measures in bonds with the same issue size, age, maturity, and credit rating category, and traded in the same month. For a description of the dependent variable and the categorical variables, please refer to Table 5 and Table 4, respectively. The measures concern only dealer-to-client trades and are aggregated on the bond-level. The sample period is 2018. Standard errors are clustered at the bond and month levels.

Swiss market exhibits more favorable price-based liquidity characteristics and lower exposure to illiquidity-related trading costs.

The results support Hypothesis 1 that corporate bonds traded in a market with publicly observable pre-trade quotes exhibit higher market quality than comparable bonds traded in a predominantly opaque OTC market. The differences in liquidity measures across the two markets are striking: Although the U.S. corporate bond market is substantially larger and exhibits higher levels of trading activity, the Swiss market displays more favorable price-based liquidity characteristics, including lower transaction costs, smaller price impacts, and greater price competitiveness. Hameed et al. (2019) argue that markets with relatively low turnover and trading activity can

nevertheless function efficiently. Consistent with this view, our findings suggest that the comparatively low trading frequency of CHF-denominated bonds does not necessarily impair market quality. These results raise the question of whether differences in market structure contribute to the observed differences in liquidity outcomes. One possible explanation is the greater degree of pre-trade transparency in the Swiss market, resulting from the exchange listing of CHF-bonds and the availability of publicly observable quotes. Prior research has argued that decentralized OTC markets may be less competitive because investors face search frictions and limited access to pricing information (e.g., Duffie et al. 2005; Yin 2005; and Abudy and Wohl 2018). In such settings, investors often negotiate bilaterally with dealers without reliable reference prices, potentially weakening their bargaining position. Our interpretation is also consistent with Chen and Zhong (2017), who show that pre-trade transparent bonds exhibit lower dispersion in bid-ask spreads and lower transaction costs. Taken together, the evidence suggests that pre-trade transparency may contribute to more competitive pricing and improved market quality, even in markets with comparatively low trading activity.

5 Differences in market structure and the benefits of pre-trade transparency

In this section, we examine the structural differences between the U.S. and Swiss corporate bond markets and discuss the potential role of pre-trade transparency in shaping market outcomes. Similar to Abudy and Wohl (2018), we argue that the observed differences in price-based liquidity measures may be related to differences in market design, namely Switzerland's pre-trade transparent exchange-based environment versus the predominantly opaque OTC structure of the U.S. market.

5.1 An opaque OTC market versus a pre-trade transparent trading environment

A key distinction between the Swiss and U.S. corporate bond markets is the central listing of bonds on a transparent CLOB in Switzerland. This structure provides market participants with publicly observable quotes and pre-trade pricing information prior to execution. By contrast, only a limited share of U.S. corporate bonds is traded on transparent exchange-based platforms. Instead, most transactions are negotiated bilaterally between investors and dealers in OTC markets, where transparency is largely restricted to post-trade reporting through the TRACE system introduced in 2002.

Table 7 reports the share of transactions executed through transparent limit order book mechanisms in the two markets. In the TRACE sample, we interpret ATS-flagged transactions as trades executed on alternative trading systems that likely employ CLOB-type protocols. Kozora et al. (2020) and the BIS Markets Committee (2016) note that many ATS platforms operate using trading mechanisms comparable to centralized exchanges. For CHF-bonds, our dataset directly identifies transactions executed in the CLOB of SIX. The table shows that 85% of all transactions in the Swiss corporate bond market were executed through the exchange order book during

Table 7 Share of trades executed through a CLOB mechanism

	U.S	Switzerland
Share of transactions traded in a CLOB (in number of trades)	28%	85%
Share traded in a CLOB (in trading volume)	6%	21%

This table presents the share of trades executed through a CLOB trading mechanism in the two markets. In the SIX-sample, these are trades executed on SIX, whereas in the TRACE-sample, we assume that such trades are executed on an ATS. The sample period is 2018. The table considers dealer-to-client and interdealer trades

2018, while only 15% were traded OTC. In contrast, only 28% of transactions in the U.S. market were executed through ATS platforms, whereas 72% occurred in OTC markets. In both countries, the volume-based share of CLOB-trades is substantially lower, indicating that larger transactions are more likely to be executed OTC. The high share of exchange-based trading in Switzerland implies a greater degree of pre-trade transparency, as quotes and orders are publicly observable to all market participants. In addition, transparency is reinforced by the obligation of dealers to route smaller client orders to the exchange. As a result, the Swiss order book combines investor limit orders with dealer quotations, creating a centralized source of executable pre-trade price information.

Figure 2 illustrates the structural differences between the U.S. and Swiss corporate bond markets by showing the share of trading volume executed through transparent order book mechanisms across trade-size categories. The figure indicates that transactions in U.S. corporate bonds are predominantly executed OTC across all trade sizes, including the smallest trades. In contrast, OTC trading is substantially less prevalent in Switzerland, particularly for smaller transactions. Retail-sized trades in CHF-denominated bonds are executed almost exclusively through the exchange order book, whereas approximately 70% of comparable trades in the U.S. market are executed OTC.

In Switzerland, the low share of OTC trading for smaller transactions is closely related to the obligation of SIX trading members—including dealers, brokers, and banks—to route these client orders to the exchange.⁹ Harris et al. (2015) argue that publicly displaying customer orders in a centralized order display facility increases competition among liquidity providers by allowing dealers to observe and interact with customer order flow. In addition, transparent limit orders provide investors with observable reference prices prior to trading, potentially strengthening the bargaining position of less informed market participants in bilateral OTC negotiations. This interpretation is consistent with Green et al. (2007), who show that dealers can exercise substantial market power when transparency is limited. Taken together, these arguments suggest that broader access to pre-trade price information may enhance price competition and contribute to lower transaction costs.

Notably, even approximately half of the volume in odd-lot transactions—which are not subject to mandatory exchange trading—is still executed through the order book in Switzerland. By contrast, trades larger than one million are conducted almost entirely OTC. The increasing reliance on OTC trading for larger transactions is consistent with the transparency-information leakage trade-off described by Hendershott

⁹ Retail orders can only be executed OTC by dealers outside the SIX continuous trading hours.

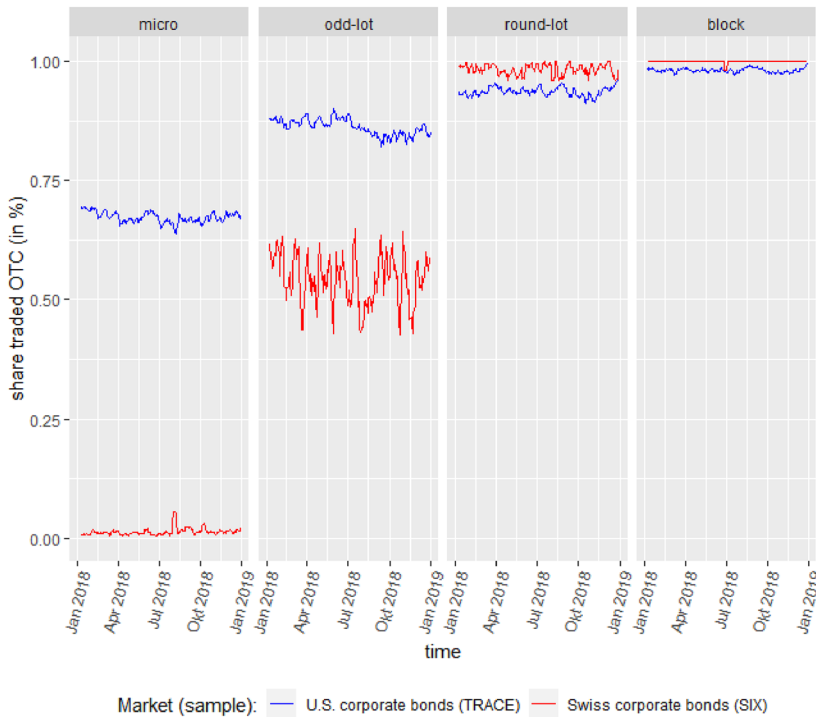


Fig. 2 Share of CLOB trading in the two markets. This figure presents time series of the share of volume traded OTC versus the share executed through a CLOB trading mechanism (5-day moving average). In the TRACE-sample, trades executed through a CLOB mechanism are defined as transactions executed on an ATS. In the SIX-sample, orders executed on the SIX order book are defined as CLOB trades. Trade size category is defined according to Hendershott and Madhavan (2015) as follows: *micro* (<100,000), *odd-lot* (100,000–1 m), *round-lot* (1–5 m), and *block* (5+ m). The charts consider dealer-to-client and interdealer trades. The sample period is 2018

and Madhavan (2015). While execution in a transparent CLOB may increase competition among liquidity providers, it can also expose traders to information leakage. Information leakage arises when other market participants observe trading intentions and adjust their price expectations accordingly, potentially increasing execution costs for large orders.

5.2 Transaction cost advantage in a pre-trade transparent market

This section examines the differences in bid-ask spreads reported in Table 5 in order to further explore the role of pre-trade transparency in shaping transaction costs. It also provides the foundation for the next section, which studies how transaction costs vary across order sizes and whether pricing differs systematically across investors. Following the approach introduced in Table 5, we calculate effective bid-ask spreads using a slightly modified version of the methodology proposed by Choi et al. (2024):

$$Spread_j = 2 \times Abs\left(\frac{TradePrice_j - BenchmarkPrice_j}{BenchmarkPrice_j}\right) \quad (3)$$

As the SIX-sample does not identify information on the trade direction (i.e., whether an investor buys or sells), we take the absolute value of the relative costs from a trade with $TradePrice_j$ compared with a benchmark price $BenchmarkPrice_j$. Furthermore, we calculate the benchmark price using the volume-weighted average price of all inter-dealer trades in a bond on a day. Unlike the statistics reported in Sect. 4, i.e., Tables 5 and 6, where effective spreads are aggregated to the bond level, the effective bid-ask spreads used in this and the following sections are calculated at the transaction level.

Figure 3 presents median bid-ask spreads in the two markets over time across different order-size categories. The figure highlights substantial differences in transaction costs between U.S. corporate bonds and CHF-denominated bonds. In particular, the Swiss market exhibits consistently lower bid-ask spreads across all trade sizes. The spread differential is most pronounced for smaller transactions and gradually narrows as order size increases. Nevertheless, lower spreads in the Swiss market persist even for larger trades, indicating that the transaction cost advantage is not limited to retail-sized orders.

Mattmann (2021) examines differences in transaction costs between electronically executed client-to-client trades on a CLOB and dealer-intermediated OTC transactions in the U.S. corporate bond market. The study documents substantially lower

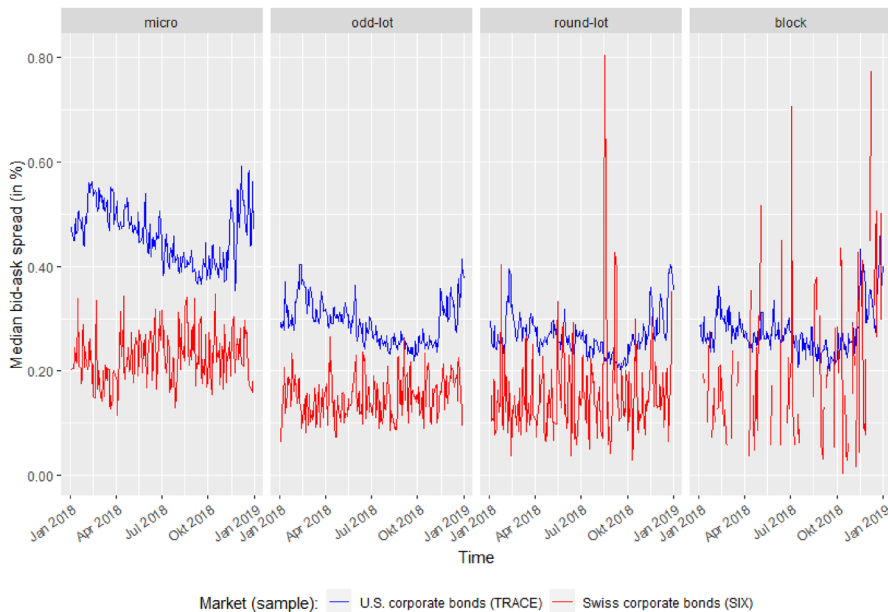


Fig. 3 Bid-ask spreads differentiated by order size. This figure presents time series of effective bid-ask spreads with order size (2-day moving average). Spreads are calculated similarly to Choi et al. (2024) using the average price of all interdealer trades. The charts only consider dealer-to-client trades. The sample period is 2018

spreads for client-to-client trades executed through the electronic order book. In particular, liquidity-demanding investors face higher costs in dealer-intermediated OTC transactions, while liquidity providers receive more favorable prices relative to comparable CLOB executions. Mattmann (2021) further shows that electronic execution with firm quotes is particularly beneficial for trades below \$1 million, including micro and odd-lot transactions. These findings are consistent with the view that transparent order book trading can reduce transaction costs by facilitating direct interaction between investors without requiring dealer intermediation. Figure 4 supports this interpretation by comparing bid-ask spreads across trading channels in the two markets. The figure shows that transactions executed through transparent order book mechanisms are associated with lower spreads than dealer-intermediated OTC trades.

The right panel of Figure 4 shows that transaction costs in the U.S. and Swiss corporate bond markets are broadly comparable for trades executed through transparent order book mechanisms. In contrast, substantial differences emerge for OTC transactions, as illustrated in the left panel. In particular, dealer-intermediated OTC trades in the U.S. market are associated with markedly wider spreads than comparable OTC trades in Switzerland. Figure 4 also shows that transaction costs in the U.S. market vary considerably across trading channels, with OTC transactions exhibiting substantially higher spreads than trades executed electronically through a CLOB. By contrast, differences across trading channels are less pronounced in Switzerland, where dealer-intermediated OTC trading appears to be significantly less costly.

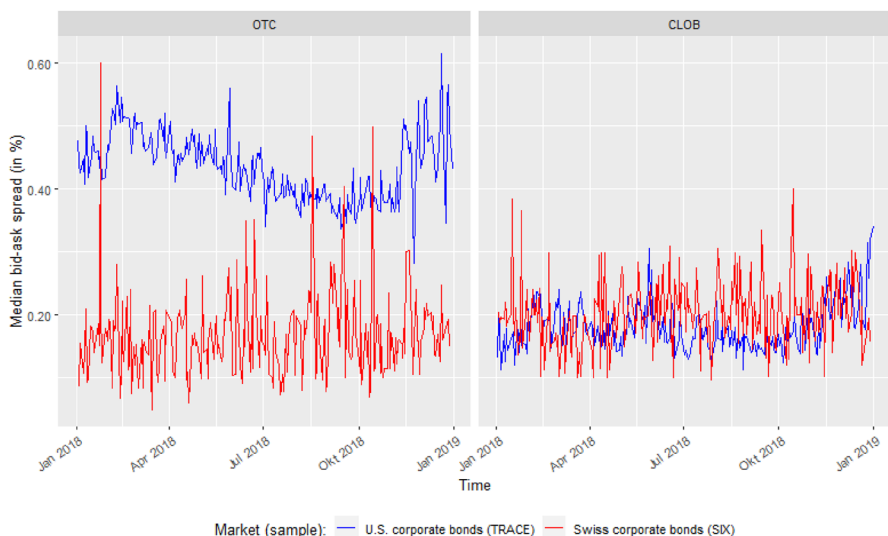


Fig. 4 Differences in bid-ask spreads separated by trade channel. This figure presents time series of effective bid-ask spreads of transactions executed OTC (left chart) and traded in a CLOB (right). In the TRACE-sample, trades executed through a CLOB mechanism are defined as transactions executed on an ATS. In the SIX-sample, orders executed on the SIX order book are defined as CLOB trades. Spreads are calculated similarly to Choi et al. (2024) using the average price of all interdealer trades. The figure only considers dealer-to-client trades. The sample period is 2018

The qualitative evidence is confirmed by the multivariate analysis reported in Table 8. Following the previous specifications, we compare trades of similar size executed in bonds with comparable issue size, age, maturity, and credit rating during the same month. The results show that the spread differences remain after controlling for these characteristics. On the one hand, Table 8 confirms that spreads for trades executed through transparent order book mechanisms are not statistically different between the U.S. and Swiss markets. This is consistent with the view that anonymous order book trading generates more standardized pricing outcomes than bilateral OTC negotiations. On the other hand, economically and statistically significant differences remain for OTC trades: spreads for OTC transactions in CHF-bonds are, on average, 0.31% points lower than for comparable U.S. corporate bond transactions.

These findings are important for two reasons. First, they suggest that the transaction cost advantage for smaller trades in the Swiss corporate bond market is closely related to the high share of transactions executed through the transparent CLOB of SIX. This is particularly evident for retail-sized trades: as shown in Figure 2, such transactions are executed almost exclusively through the exchange order book in Switzerland, whereas approximately 70% of comparable trades in the U.S. market are executed OTC. Accordingly, the spread differences for smaller trades appear closely linked to the greater reliance on dealer intermediation in the U.S. market. Second, the results indicate that the Swiss market also differs in the pricing of OTC transactions. Even in dealer-intermediated OTC trading, transaction costs are substantially lower in Switzerland than in the U.S. market. This raises the question of whether the transparent exchange environment in Switzerland may also influence pricing outcomes in decentralized OTC trading.

5.3 Association between a pre-trade transparent CLOB and OTC trading

An important question is why publicly observable CLOB quotes provide useful information for institutional OTC transactions, which are typically substantially larger than exchange trades. Following Kyle (1985), larger trades are expected to command additional compensation for inventory risk and price impact, implying that the execu-

Table 8 Differences in spreads in the two markets differentiated by trade channel

	CLOB	OTC
CHF-bonds	-0.0119 (0.0096)	-0.3051*** (0.0115)
<i>Fixed effect:</i>		
Order size-issue size-age-maturity-rating-month FE	Yes	Yes
Observations	46,016	362,423
R ²	0.21	0.23

This table estimates spreads according to Eq. (3) for orders traded in a CLOB and executed OTC. The regression specification follows Eq. (1). The variable *CHF-bonds* is a market dummy for bonds in the Swiss market (the SIX-sample) and indicates differences in the liquidity measures compared with the U.S. market (the TRACE-sample), thus capturing differences in market design. *Order size-issue size-age-maturity-rating-month fixed effect* controls for spreads in trades with the same order size, for bonds with the same issue size, age, maturity, and credit rating category, and traded in the same month. For a description of the independent variables, please refer to Table 4. The table concerns only dealer-to-client trades. The sample period is 2018. Standard errors are clustered at the bond and month levels

tion price of a large OTC transaction will generally differ from the quoted price of a small CLOB trade. We therefore do not argue that CLOB quotes directly determine the execution price of large OTC transactions. Instead, we argue that CLOB quotes provide a transparent and continuously observable reference price or anchor that reflects prevailing market conditions for both parties of a transaction before negotiations begin. Dealers and investors may subsequently negotiate a trade-size-specific premium, but these negotiations begin from a commonly observable reference price rather than in an informational vacuum. Consequently, while the absolute execution price of a large OTC transaction may differ from the quoted CLOB price, the existence of publicly observable pre-trade quotes may nevertheless improve bargaining outcomes and reduce transaction costs. Similar to Harris (2015), we argue that the presence of pre-trade information for investors in Switzerland, supplied by the exchange-listings of CHF-bonds, provides strong incentives for dealers to offer competitive prices in OTC trading.

This interpretation is further supported by the results in Table 9, which examines the relationship between transaction size and bid-ask spreads. The table reports spread estimates for all transactions as well as for OTC trades only. Following the previous specifications, we include fixed effects to compare transactions executed in the same bond, on the same day, and through the same trading channel (OTC versus CLOB). Consistent with earlier studies such as Schultz (2001) and Edwards et al. (2007), the results reveal a strong negative relationship between trade size and transaction costs in the U.S. corporate bond market. Smaller trades incur substantially higher spreads than larger transactions, and this pattern remains robust across different subsamples. For example, when restricting the analysis to OTC transactions, spreads for odd-lot, round-lot, and block trades are 30, 46, and 45 basis points lower, respectively, than spreads for micro-sized retail trades. Similarly, when retail-sized trades are excluded, round-lot and block trades continue to exhibit significantly lower spreads than odd-lot transactions. These findings indicate that transaction size is an important determinant of trading costs in dealer-intermediated U.S. corporate bond trading, with smaller investors facing systematically less favorable pricing. Theoretical work by Duffie et al. (2005) suggests that such patterns may reflect differences in bargaining power and informational asymmetries in bilateral OTC negotiations.

The negative relationship between transaction size and spreads is also present in the Swiss corporate bond market, although it is substantially weaker than in the U.S. market. For example, across all transactions, odd-lot trades exhibit spreads that are only 5 basis points lower, on average, than spreads for micro-sized retail trades. For OTC transactions, however, spreads do not differ significantly across order-size categories. These findings suggest that dealers in the Swiss market are less able to charge systematically higher spreads to smaller and potentially less informed investors in bilateral OTC negotiations. One possible explanation is the greater degree of pre-trade transparency provided by the exchange-listed environment. Investors trading CHF-denominated bonds OTC can observe firm quotations in the CLOB of SIX prior to negotiation, allowing them to use publicly observable bid and ask prices as reference points when interacting with dealers. Because these prices are visible to all market participants, dealers may face stronger incentives to offer competitive quotes in OTC trading. In addition, the centralized listing of bonds on SIX likely

Table 9 Transaction cost estimates for trades in the U.S. and Swiss bond markets

	SIX-sample		TRACE-sample			
	All trades		OTC trades		All trades	
	(OTC and CLOB)		(excl. retail)	(OTC and CLOB)		
Odd-lot	-0.0514** (0.0170)	-0.2983 (0.2259)		-0.2966*** (0.0014)	-0.2970*** (0.0014)	OTC trades (excl. retail)
Round-lot	-0.0508. (0.0296)	-0.2819 (0.2180)	0.0080 (0.0253)	-0.4606*** (0.0023)	-0.4610*** (0.0023)	-0.1500*** (0.0018)
Block	-0.0375 (0.0307)	-0.2710 (0.2185)	0.0274 (0.0254)	-0.4536*** (0.0029)	-0.4536*** (0.0029)	-0.1291*** (0.0023)
<i>Fixed effect:</i>						
Bond-day-trade channel FE	Yes	No	No	Yes	No	No
Bond-day FE	No	Yes	Yes	No	Yes	Yes
Observations	59,208	9,115	8,742	6,587,209	6,496,206	1,907,889
R ²	0.68	0.75	0.80	0.54	0.53	0.70

This table estimates spreads according to Eq. (3) for CHF-bonds (SIX) and U.S. corporate bonds (TRACE). The spreads are calculated using Eq. (1). The trade size categories are defined according to Hendershott and Madhavan (2015) and are described in Table 4. CLOB trades are orders traded in the order book of the exchange (the SIX-sample) or transacted through an ATS platform (the TRACE-sample). The fixed effect variable *Bond-day-trade channel FE* refers to orders in the same bond, on the same day, and executed in the same trade channel (OTC vs. CLOB trades). The table only considers dealer-to-client trades. The sample period is 2018. Standard errors are clustered at the bond and day levels

reduces investors' search costs when evaluating dealer quotes. Yin (2005) shows that even small search costs can materially affect dealer pricing behavior and market outcomes. The study argues that centralized and publicly available pricing information may induce dealers to quote more competitive prices and reduce price discrimination across investors. Our findings are also consistent with Abudy and Wohl (2018), who document limited price discrimination in the centralized Israeli corporate bond market. However, unlike the present study, they are unable to examine whether such effects extend to OTC trading because nearly all Israeli corporate bond transactions are executed through a centralized order book mechanism.

Overall, the results presented in Sects. 5.1–5.3 support Hypothesis 2, namely that the improved market quality associated with pre-trade transparency extends beyond CLOB trading and translates into OTC transactions. The evidence indicates that the differences in market quality documented between the Swiss and U.S. corporate bond markets are largely confined to the OTC trading channel, whereas transaction costs for trades executed through transparent order book mechanisms are remarkably similar across the two markets. These findings suggest that the benefits of pre-trade transparency arise not only within the CLOB itself but also through spillover effects on OTC trading.

5.4 Firmly quoted bid-ask prices used as benchmarks in OTC negotiations

Our interpretation that pre-trade transparency may influence OTC pricing is further supported by Figure 5. Because our dataset contains quote data for all CHF-bonds listed on SIX, we are able to compare OTC transaction prices with the average daily bid and ask quotes displayed in the exchange order book on the day of execution. This comparison allows us to examine the extent to which prices negotiated in bilateral OTC trading relate to publicly observable quotes in the CLOB. More specifically, it provides evidence on whether quoted bid and ask prices in the exchange order book serve as reference points or benchmark boundaries for OTC negotiations.

Figure 5 shows that 95% of OTC transactions in CHF-denominated bonds are executed within the average daily bid-ask range quoted in the CLOB of SIX. Moreover, 63% of OTC transaction prices are closer to the quoted midpoint than to either the bid or ask quote. These findings suggest that publicly observable quotes in the exchange order book serve as economically meaningful reference points for bilateral OTC negotiations. This supports Hypothesis 3, which predicts that OTC transaction prices largely remain within the average daily CLOB bid-ask range observed on the

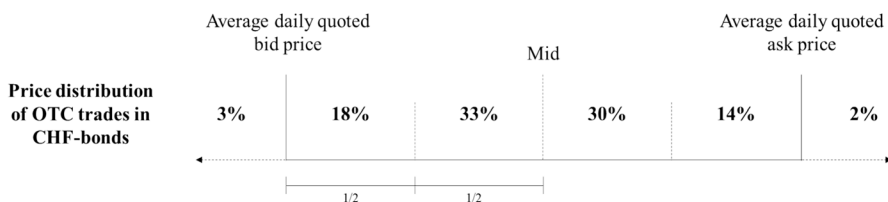


Fig. 5 Price distribution of OTC trades relative to the prices quoted in the CLOB. This figure presents the price distribution of OTC transactions in CHF-bonds relative to the average daily bid-ask prices quoted in the SIX CLOB. The figure only considers dealer-to-client orders executed OTC in 2018

execution date. The quotes in the CLOB appear to define a benchmark range within which investors and dealers negotiate transaction prices, while many executions occur at prices closer to the midpoint than to the displayed quote boundaries. Only 5% of OTC transactions occur outside the quoted spread range. The evidence further suggests that the transparent order book may influence OTC trading outcomes in addition to providing an efficient execution venue for smaller transactions. By making executable quotes publicly observable, the exchange environment may improve investors' outside options and reduce informational asymmetries in dealer negotiations. Craig et al. (2020) argue that access to pre-trade information lowers informational advantages in dealer markets and may contribute to lower transaction costs. Importantly, these patterns are not limited to retail-sized trades. The findings are also consistent with the evidence in Chen and Zhong (2017), who show that institutional investors face lower bid-ask spreads when trading NYSE-listed bonds. They argue that pre-trade transparency enhances investors' bargaining positions relative to dealers. Our results are also in line with the theoretical framework of Duffie et al. (2017), who show that benchmark prices can mitigate informational frictions and improve matching efficiency in decentralized OTC markets.

The following analysis provides additional insights into the pricing and liquidity conditions in the SIX order book. Figure 6 examines the distribution of quoted and effective bid-ask spreads for CHF-denominated corporate bonds. The left panel reports the cumulative distribution of median quoted spreads, while the right panel presents the cumulative distribution of implied effective spreads based on executed transactions. To estimate effective spreads, we apply Eq. (3) using the quoted midpoint from the order book as the benchmark price for trades executed in the CLOB. Effective spreads are then obtained by doubling the deviation of transaction prices from the quoted midpoint. Comparing quoted and effective spreads provides information on the extent to which executed trades improve upon quoted boundaries.

The left panel of Figure 6 shows that 50% of CHF-denominated bonds are quoted with spreads of 0.56% or less in the SIX order book. Moreover, quoted spreads

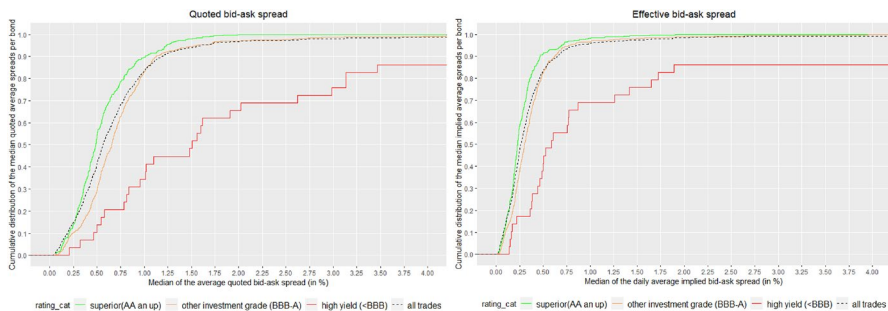


Fig. 6 Median of the average quoted and implied daily bid-ask spreads. This figure presents the cumulative distribution function of the median quoted (left chart) and implied (right) average spreads per CHF-bond. Quoted spreads are defined as the difference between the average quoted bid and ask price relative to its quoted mid price. The implied or effective spreads are calculated according to Eq. (3), but using the average quoted mid-price as the reference price. For the implied spreads, only dealer-to-client trades are considered. To receive the median quoted and implied spread per bond, we take the median of the average daily quoted/effective spreads per bond over the sample period of 2018

remain below 1% for approximately 80% of all bonds. Because all CHF-bonds in the sample are listed on SIX, the figure provides a comprehensive picture of quoted liquidity conditions in the Swiss corporate bond market. The figure also indicates substantial differences across credit quality categories. Higher-rated bonds tend to exhibit narrower quoted spreads, whereas high-yield bonds face significantly wider spreads. This pattern is consistent with Hendershott and Madhavan (2015), who argue that liquidity providers in quote-driven markets are exposed to adverse selection risk because firm bid and ask quotes effectively provide free trading options to market participants. These risks are likely more pronounced for riskier and less actively traded securities, such as high-yield bonds. Related theoretical work by Copeland and Galai (1983) interprets limit orders as implicit options granted to other market participants. Barclay et al. (2006) further argue that the value of these implicit options increases for less liquid securities because limit orders remain exposed to price fluctuations for longer periods within the order book.

The right panel of Figure 6 shows that implied effective spreads are substantially lower than quoted spreads. For example, 50% of CHF-denominated bonds exhibit a median implied spread below 0.26%, while approximately 80% record implied spreads of 0.47% or less. Relative to quoted spreads, these execution costs are lower by more than one-half on average. These findings suggest that transactions in the CLOB are frequently executed at prices within the quoted bid-ask range rather than directly at the displayed bid or ask quote. In other words, market participants often receive price improvements relative to the quoted spread boundaries. One possible explanation is that investors place limit orders within the prevailing spread, allowing trades to occur at prices closer to the midpoint of the order book. This mechanism may be particularly beneficial for smaller investors because it enables them to interact directly with other market participants through a centralized and publicly observable trading venue. The evidence is consistent with Barclay et al. (1999), who argue that the incorporation of limit orders into Nasdaq improved execution quality by increasing competition within the trading process. Similarly, Harris et al. (2015) suggest that introducing a public order display facility in the U.S. corporate bond market could encourage greater use of limit orders by allowing investors to interact directly with other clients and dealers through transparent quotations.

Figure 7 complements the analysis of bid-ask spreads by examining quoted depth in the SIX order book, measured as the maximum quantity available at the best bid and ask quotes. The figure reports the cumulative distribution of average daily quoted depth per bond on both sides of the order book. The results indicate substantial displayed liquidity in the Swiss market. Approximately 79% (76%) of CHF-bonds exhibit quoted depth of at least 100,000 on the bid (ask) side. This suggests that the order book is capable of immediately absorbing most retail-sized transactions at the best displayed quotes. Moreover, around 60% of bonds are quoted with available depth between 100,000 and 200,000. The figure also reveals differences across credit quality categories. Higher-rated bonds tend to exhibit greater quoted depth, whereas lower-rated bonds are associated with lower immediately available quantities.

These quoted volumes are economically meaningful relative to observed trading activity. The average (median) trade size executed in the SIX order book is 46,000 (25,000), while the average (median) daily quoted volume amounts to 144,000

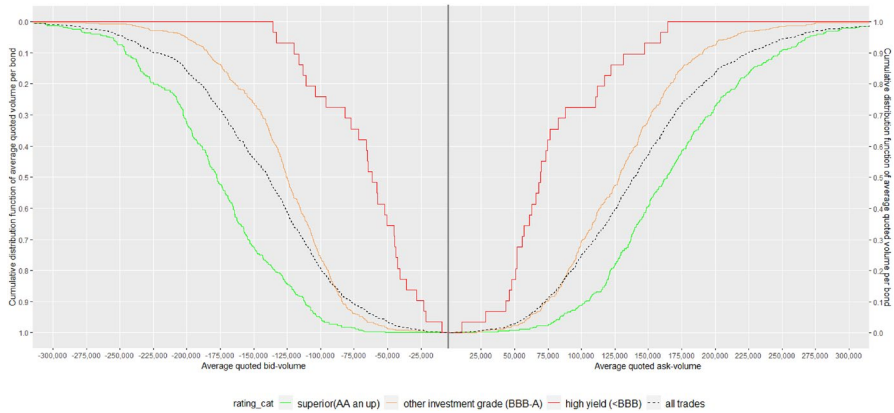


Fig. 7 Average quoted daily bid-ask volume of CHF-bonds. This figure presents the cumulative distribution of the average quoted bid and ask volume per CHF-bond. The average volume is calculated as the average of the quoted par value times the price. To receive the average quoted volume per bond, we take the mean of the daily average quoted volume per bond over the sample period of 2018

(113,000). Taken together, these figures indicate that displayed liquidity in the SIX order book is generally sufficient to accommodate a substantial share of retail-sized transactions. The evidence is also informative when viewed relative to the U.S. market. Given the median trade size of approximately \$35,000 in U.S. corporate bonds reported in Table 2, the displayed depth in the SIX order book appears capable of accommodating a meaningful portion of typical retail-sized transactions observed in the U.S. market as well.

The descriptive evidence in Figures 6 and 7 indicates substantial cross-sectional variation in the quality of publicly displayed CLOB quotes. Some bonds exhibit relatively narrow quoted bid-ask spreads and substantial displayed volumes, whereas others are characterized by wider quotes and lower available quantities. If CLOB quotes provide meaningful reference prices for OTC negotiations, bonds with more competitive and deeper displayed quotes in the CLOB should also exhibit more favorable pricing in the OTC market. To examine this prediction, we relate OTC effective bid-ask spreads to two bond-level measures of CLOB quote quality. The first measure is the quoted CLOB spread, defined as the difference between the average daily bid and ask prices relative to the average quoted midpoint. The second measure is quoted depth, calculated as the average of the quoted bid and ask volume, where quoted volume is measured as the quoted par value multiplied by the corresponding average daily price. For each bond-day, we calculate the average effective bid-ask spread of dealer-to-client OTC transactions and match it with the CLOB quote measures observed for the same bond and trading day. Table 10 reports the regression results. Columns (1) and (2) relate OTC effective bid-ask spreads to the quoted CLOB spread, while Columns (3) and (4) use quoted depth as the explanatory variable. Columns (5) and (6) include both measures simultaneously.

Across all specifications, the quoted CLOB spread is positively and significantly associated with OTC effective bid-ask spreads. Bonds characterized by wider quoted bid-ask spreads in the CLOB also exhibit higher transaction costs in the OTC market.

Table 10 Association between CLOB quote quality and OTC effective bid-ask spreads

	OTC spreads	OTC spreads	OTC spreads	OTC spreads	OTC spreads	OTC spreads
	(1)	(2)	(3)	(4)	(5)	(6)
Quoted CLOB spread	0.3697*** (0.0480)	0.3565*** (0.0687)			0.3462*** (0.0454)	0.3420*** (0.0680)
Log quoted depth			– 0.5727** (0.1746)	– 0.3078*** (0.0749)	– 0.158* (0.0735)	– 0.1221* (0.0569)
Bond controls	Yes	No	Yes	No	Yes	No
<i>Fixed effect:</i>						
Month	Yes	No	Yes	No	Yes	No
Size-age-maturity-rating-month FE	No	Yes	No	Yes	No	Yes
Observations	5,373	5,213	5,373	5,213	5,373	5,213
R2	0.45	0.56	0.22	0.47	0.45	0.57

This table presents the association between the quality of publicly observable CLOB quotes and OTC transaction costs for Swiss corporate bonds. The dependent variable is the average daily effective bid-ask spread of dealer-to-client OTC transactions for a given bond. The quoted CLOB spread and quoted depth are measured for the same bond and trading day. Quoted spread is defined as the difference between the average daily ask and bid prices relative to the average daily quoted midpoint. Quoted depth is calculated as the average of the quoted bid and ask volume, where quoted volume is measured as the quoted par value multiplied by the corresponding average daily price. Quoted depth enters the regressions in logarithmic form. Columns (1), (3), and (5) include month fixed effects and categorical controls for *issue size*, *age*, *maturity*, and *rating*. Columns (2), (4), and (6) include interacted *issue size-age-maturity-rating-month fixed effects*. The sample period is 2018. Standard errors are clustered at the bond level

The estimated coefficient remains stable after controlling for bond characteristics or, alternatively, including fixed effects. Quoted depth exhibits the opposite relation: Bonds with greater displayed volume in the CLOB tend to exhibit lower OTC effective bid-ask spreads. When quoted spread and quoted depth are included jointly, the coefficient on quoted spread remains significant, whereas the coefficient on quoted depth declines in magnitude and is only weakly significant, suggesting that quoted spreads capture most of the information relevant for OTC pricing.

These findings are economically important because they indicate that publicly observable CLOB quotes contain information that is reflected in bilateral OTC negotiations. While Figure 5 shows that most OTC transaction prices lie within the average daily bid-ask range quoted in the CLOB, the regressions demonstrate that differences in the quality of publicly displayed quotes across bonds are associated with differences in OTC transaction costs. Bonds characterized by tighter quoted spreads and greater displayed volume also exhibit lower effective bid-ask spreads in the OTC market.

This evidence directly addresses the question of whether quotes displayed in a transparent order book are informative beyond transactions executed on the exchange itself. Larger transactions are typically negotiated bilaterally in the OTC market and may exceed the quantities immediately available in the CLOB. Nevertheless, the results suggest that publicly observable quotes provide economically meaningful reference prices for these negotiations. Even when investors ultimately execute transactions OTC, dealers are aware that quoted prices can be benchmarked against

contemporaneous executable quotes displayed in the order book. Consistent with the bargaining mechanism proposed by Chen and Zhong (2017), publicly observable pre-trade information may therefore strengthen investors' bargaining position and increase competitive pressure among dealers.

Although the analysis is based on cross-sectional associations and does not establish a causal relationship, it provides additional evidence consistent with the interpretation that pre-trade transparency generates informational benefits beyond the execution of trades in the CLOB itself.

6 Conclusion

This paper studies the relationship between pre-trade transparency and market quality in corporate bond markets. We compare the predominantly OTC-driven U.S. corporate bond market with the Swiss corporate bond market, where publicly observable executable quotes are available through a CLOB while an OTC market continues to coexist for larger transactions. This institutional setting allows us to examine whether broader access to pre-trade pricing information is associated with differences in liquidity and trading outcomes. Using transaction-level data from the Swiss market and the U.S. TRACE database, we document several robust findings. Although the Swiss corporate bond market is substantially smaller and exhibits lower trading activity, it displays significantly more favorable price-based liquidity characteristics. Transaction costs, price dispersion, and price impact are all lower in Switzerland than in the U.S. These differences persist after controlling for observable bond characteristics such as issue size, maturity, age, and credit quality.

Importantly, the differences are not confined to exchange-based trading: OTC transactions in Switzerland are also executed at substantially more competitive prices than comparable OTC transactions in the U.S. market. Moreover, OTC transaction prices in Switzerland closely track publicly observable quotes displayed in the exchange order book. The vast majority of OTC transactions occur within quoted bid-ask boundaries, and a large share of trades are even executed closer to quoted midpoint prices. These findings suggest that publicly observable quotes may serve as economically meaningful benchmarks in decentralized dealer negotiations. This interpretation is further supported by the finding that the quality of publicly observable CLOB quotes is associated with OTC transaction costs. Bonds with narrower quoted bid-ask spreads and greater quoted volume in the CLOB also exhibit lower effective bid-ask spreads in OTC trading, even after controlling for bond characteristics. These results provide additional evidence that publicly observable executable quotes contain economically meaningful information for bilateral dealer negotiations and may improve OTC pricing by serving as common reference points for market participants. We interpret the evidence as being consistent with the view that pre-trade transparency improves investors' outside options and bargaining positions in OTC markets. When executable quotes are publicly observable prior to trading, dealers may face stronger competitive pressure and reduced ability to extract informational rents from less informed investors. In this sense, the benefits of transparency

may extend beyond centralized trading venues and influence pricing outcomes even in bilateral OTC environments.

Our findings contribute to the growing literature on corporate bond market structure and pre-trade transparency. While prior research has largely focused on the effects of post-trade transparency, our results highlight the potential importance of publicly available pre-trade pricing information. More broadly, the paper suggests that market design may play an important role in shaping trading costs and price competitiveness in decentralized financial markets. At the same time, our analysis also highlights important tradeoffs associated with transparent trading environments. Larger transactions continue to be executed predominantly OTC in both markets, consistent with concerns regarding information leakage and execution risk in transparent order books. The results therefore do not imply that fully centralized exchange trading is optimal for all corporate bond transactions. Rather, they suggest that broader access to publicly observable pre-trade prices may improve market quality even when dealer intermediation remains an important component of the trading process.

Several directions for future research remain. In particular, the removal of mandatory exchange routing requirements for smaller trades in Switzerland after 2018 offers a promising opportunity to study how reductions in trading obligation affect trading behavior and OTC pricing outcomes within the same market structure. More generally, future work could further examine how pre-trade transparency interacts with electronic trading protocols.

Acknowledgements I thank the editors and an anonymous referee for their helpful comments and suggestions. The paper has also improved thanks to comments from Adrian Knoblauch, Avi Wohl, David Krein, Hanspeter Haag, John Berchin, Jürg Fausch, Marco Wirz, Norman Schürhoff, Pascal Gantenbein, Philippe Oster, Sajiv D'Cunja, Tim A. Kroencke, and Ueli Goldener. We thank SIX Swiss Exchange for providing part of the data used in this paper and for valuable feedback at various presentations.

Author contributions This is a single-authored paper.

Funding Open access funding provided by Lucerne University of Applied Sciences and Arts

Data availability The data supporting the findings of this study are proprietary and were provided by SIX Swiss Exchange; accordingly, they are not publicly available. The data may, however, be obtained from the author upon reasonable request and subject to permission from SIX Swiss Exchange. Data for the U.S. market (TRACE) were accessed through Wharton Research Data Services (WRDS).

Declarations

Conflict of interest The author declares no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Abudy, M.M., Wohl, A.: Corporate bond trading on a limit order book exchange. *Rev. Finance* **22**, 1413–1440 (2018)
- Allen, J., Wittwer, M.: Centralizing over-the-counter markets? *J. Polit. Econ.* **131**, 3310–3351 (2023)
- Amihud, Y.: Illiquidity and stock returns: cross-section and time-series effects. *J. Finance Mark.* **5**, 31–56 (2002)
- Anand, A., Jotikasthira, C., Venkataraman, K.: Mutual fund trading style and bond market fragility. *Rev. Finance Stud.* **34**, 2993–3044 (2021)
- BIS Markets Committee: Electronic trading in fixed income markets, Markets Committee Publications, 7, 21 January (2016)
- Barclay, M.J., Christie, W.G., Harris, J.H., Kandel, E., Schultz, P.H.: Effects of market reform on the trading costs and depths of Nasdaq stocks. *J. Finance* **54**, 1–34 (1999)
- Barclay, M.J., Hendershott, T., Kotz, K.: Automation versus intermediation: evidence from treasuries going off the run. *J. Finance* **61**, 2395–2414 (2006)
- Bessembinder, H., Jacobsen, S., Maxwell, W., Venkataraman, K.: Capital commitment and illiquidity in corporate bonds. *J. Finance* **73**, 1615–1661 (2018)
- Bessembinder, H., Maxwell, W., Venkataraman, K.: Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *J. Finance Econ.* **82**, 251–288 (2006)
- Biais, B., Green, R.: The microstructure of the bond market in the 20th century. *Rev. Econ. Dyn.* **33**, 250–271 (2019)
- Boehmer, E., Saar, G., Yu, L.: Lifting the veil: an analysis of pre-trade transparency at the NYSE. *J. Finance* **60**, 783–815 (2005)
- Chen, F., Zhong, Z.: Pre-trade transparency in over-the-counter bond markets. *Pac. Basin Finance J.* **45**, 14–33 (2017)
- Choi, J., Huh, Y., Shin, S.S.: Customer liquidity provision: implications for corporate bond transaction costs. *Manag. Sci.* **70**, 187–206 (2024)
- Copeland, T.E., Galai, D.: Information effects on the bid-ask spread. *J. Finance* **38**, 1457–1469 (1983)
- Craig, L., Harris, L., Shohfi, T.: Riskless principal trades in corporate bond markets. *J. Fixed Income* **35**, 18–29 (2025)
- Craig, L., Kim, A., & Woo, S. W.: Pre-trade information in the municipal bond market. *Division of Economic and Risk Analysis of the SEC* (2020)
- de Roure, C., Moench, E., Pelizzon, L., Schneider, M.: OTC discount, *Management Science*, forthcoming. (2025)
- Dick-Nielsen, J., Feldhütter, P., Lando, D.: Corporate bond liquidity before and after the onset of the subprime crisis. *J. Finance Econ.* **103**, 471–492 (2012)
- Duffie, D., Dworczak, P., Zhu, H.: Benchmarks in search markets. *J. Finance* **72**, 1983–2044 (2017)
- Duffie, D., Garleanu, N., Pedersen, L.H.: Over-the-counter markets. *Econometrica* **73**, 1815–1847 (2005)
- Edwards, A.K., Harris, L.E., Piwowar, M.S.: Corporate bond market transaction costs and transparency. *J. Finance* **62**, 1421–1451 (2007)
- Glode, V., Opp, C.C.: Over-the-counter versus limit-order markets: the role of traders' expertise. *Rev. Financ. Stud.* **33**, 866–915 (2020)
- Goldstein, M.A., Hotchkiss, E.S.: Providing liquidity in an illiquid market: dealer behavior in US corporate bonds. *J. Finance Econ.* **135**, 16–40 (2020)
- Goldstein, M.A., Hotchkiss, E.S., Sirri, E.R.: Transparency and liquidity: a controlled experiment on corporate bonds. *Rev. Finance Stud.* **20**, 235–273 (2007)
- Green, R.C., Hollifield, B., Schürhoff, N.: Financial intermediation and the costs of trading in an opaque market. *Rev. Finance Stud.* **20**, 275–314 (2007)
- Hameed, A., Helwege, J., Li, R., & Packer, F.: Measuring corporate bond liquidity in emerging market economies: price-vs quantity-based measures. BIS paper no. 102e (2019)
- Harris, L., Kyle, A.S., Sirri, E.R.: Statement of the Financial Economists Roundtable, April 2015: the structure of trading in bond markets. *Finance Anal. J.* **71**, 5–8 (2015)
- Harris, L.E., Piwowar, M.S.: Secondary trading costs in the municipal bond market. *J. Finance* **61**, 1361–1397 (2006)
- Harris, L.: Transaction costs, trade throughs, and riskless principal trading in corporate bond markets, Working paper, University of Southern California (2015)

- Hendershott, T., Li, D., Livdan, D., Schürhoff, N.: Relationship trading in over-the-counter markets. *J. Finance* **75**, 683–734 (2020)
- Hendershott, T., Madhavan, A.: Click or call? Auction versus search in the over-the-counter market. *J. Finance* **70**, 419–447 (2015)
- Hendershott, T. J., Livdan, D., Schürhoff, N.: All-to-all liquidity in corporate bonds, Working paper, Swiss Finance Institute, Research Paper Series, No. 21–43 (2021)
- Holden, C. W., Liu, X., Mao, Y., & Nam, J.: Price discovery in the stock, otc corporate bond, and nyse corporate bond markets. Kelley School of Business Research Paper, (18-53) (2018)
- Hong, G., Warga, A.: An empirical study of bond market transactions. *Finance Anal. J.* **56**, 32–46 (2000)
- Hotchkiss, E., Jostova, G.: Determinants of corporate bond trading: a comprehensive analysis. *Q. J. Finance* **07**, 1750003 (2017)
- Knoblauch, Adrian P.: Fokus Fixed Income: CHF-Kapitalmarkt für Anleihen - Bilder sagen mehr als tausend Worte, Studie der Zürcher Kantonalbank, 18. August (2021)
- Kozora, M., Mizrach, B., Peppe, M., Or Shachar, Sokobin, J.: Alternative trading systems in the corporate bond market, Federal Reserve Bank of New York, Staff Report No. 938 (2020)
- Kyle, A.S.: Continuous auctions and insider trading. *Econometrica* **53**, 1315–1335 (1985)
- Lee, T., Wang, C.: Regulating over-the-counter markets. *J. Finance* **80**, 1929–1962 (2025)
- Mattmann, B.: Customer Liquidity Provision in Corporate Bond Markets: Electronic Trading Versus Dealer Intermediation, Working paper, University of Basel (2021)
- Odegaard, B.A.: Bond liquidity at the oslo stock exchange, Staff Memo, No 5, Norges Bank (2017)
- O'Hara, M., Wang, Y., Zhou, X.: The execution quality of corporate bonds. *J. Finance Econ.* **130**, 308–326 (2018)
- O'Hara, M., Zhou, X.: Anatomy of a liquidity crisis: corporate bonds in the COVID-19 crisis. *J. Finance Econ.* **142**, 46–68 (2021a)
- O'Hara, M., Zhou, X.: The electronic evolution of corporate bond dealers. *J. Finance Econ.* **140**, 368–390 (2021b)
- Plante, S.: Should corporate bond trading be centralized?, Working paper, University of Pennsylvania (2017)
- Randall, O.: How do inventory costs affect dealer behavior in the us corporate bond market?, Working paper, Emory University (2015)
- Schultz, P.: Corporate bond trading costs: a peek behind the curtain. *J. Finance* **56**, 677–698 (2001)
- Schultz, P.: Inventory management by corporate bond dealers, Working paper, University of Notre Dame. (2017)
- U.S. Securities and Exchange Commission: Report to congress on access to capital and market liquidity, August. (2017)
- Weston, J.P.: Competition on the Nasdaq and the impact of recent market reforms. *J. Finance* **55**, 2565–2598 (2000)
- Yin, X.: A comparison of centralized and fragmented markets with costly search. *J. Finance* **60**, 1567–1590 (2005)
- Zitzewitz, E.: Paired corporate bond trades, Working paper, Dartmouth College. (2010)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Brian Mattmann is a Lecturer at the Lucerne University of Applied Sciences and Arts. He holds a Ph.D. in Finance from the University of Basel, and his research focuses on asset management and financial markets. He has professional experience in equity research, portfolio management, and fixed-income and derivatives trading. In addition to his academic activities, he holds board positions in the asset management industry.